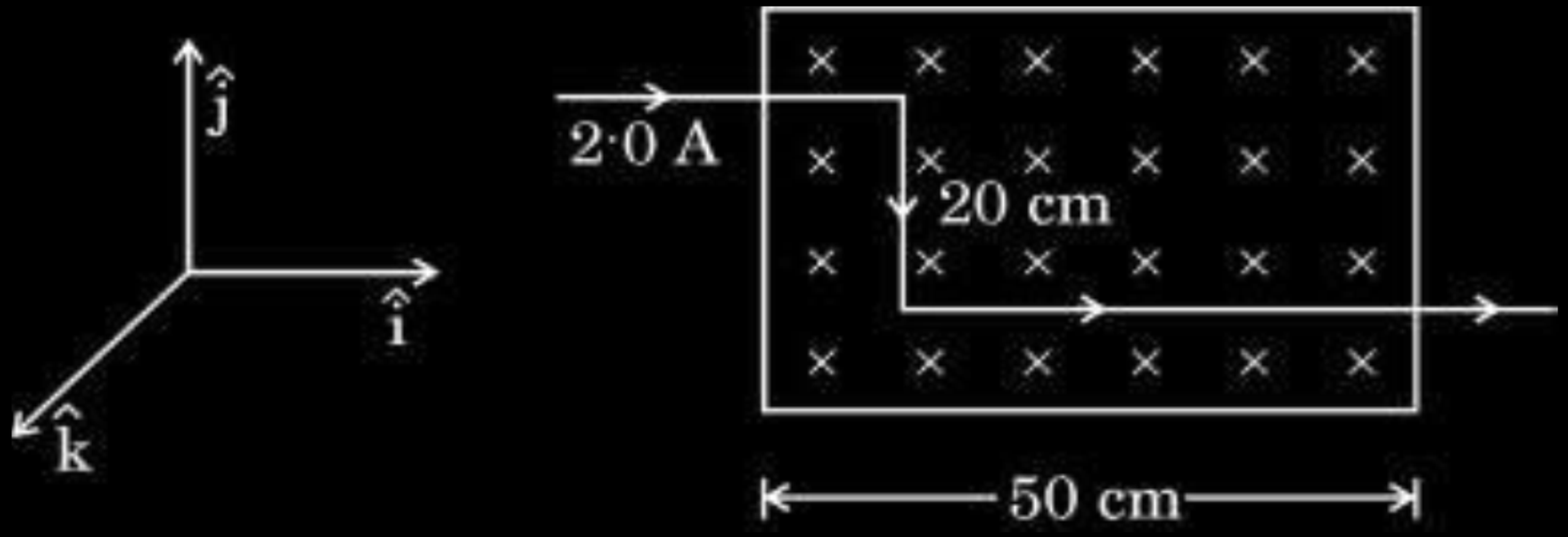
(b) (i) Derive an expression for the force $\vec{F}$ acting on a conductor of length L and area of cross-section A carrying current I and placed in a magnetic field $\vec{B}$.

(ii) A part of a wire carrying 2·0 A current and bent at 90° at two points is placed in a region of uniform magnetic field $\vec{B} = -(0{\cdot}50\ T)\ \hat{k}$, as shown in the figure. Calculate the magnitude of the net force acting on the wire. CBSE 2026 5



(i) Let the number density of the free electron in the conductor be n. So, the total charge of conductor,	
$q = nALe$	½
Total force acting on q placed in field B, moving with velocity v_d	
$\vec{F} = q(\vec{v} \times \vec{B})$	½
$\vec{F} = n(AL)e(\vec{v_d} \times \vec{B})$	1
Here, $ne\ \vec{v_d}$ is current density $(\vec{J})$ and $\lvert ne\ \vec{v_d} \rvert A$ is current in the conductor, Therefore	
$\vec{F} = [\vec{J}(AL)] \times \vec{B}$	
$\vec{F} = I(\vec{L} \times \vec{B})$	½
(ii) The force acting on the wire l_1 (50cm) placed along x-axis	
$\vec{F_1} = I(\vec{l_1} \times \vec{B})$	½
$= 2(50 \times 10^{-2})\,\hat{\imath} \times (-0.5))\,\hat{k}$	½
$= 0.5\hat{\jmath}$ N	
The force acting on the wire l_2 (20cm) placed along y-axis	
$\vec{F_2} = I(\vec{l_2} \times \vec{B})$	
$= 2(20 \times 10^{-2})(\hat{\jmath}) \times (-0.5)\,\hat{k}$	
$= 0.2\,\hat{\imath}$ N	½
As F_1 and F_2 are perpendicular to each other, the resultant force on the wire will be	
$F = \sqrt{(F_1^2 + F_2^2)}$	½
$= \sqrt{(0.25 + 0.04)}$	
$= \sqrt{0.29}N$	

Alternatively,

$$\vec{F} = \vec{F_1} + \vec{F_2}$$

$$\vec{F} = I(\vec{l_x} \times \vec{B}) + I(\vec{l_y} \times \vec{B})$$

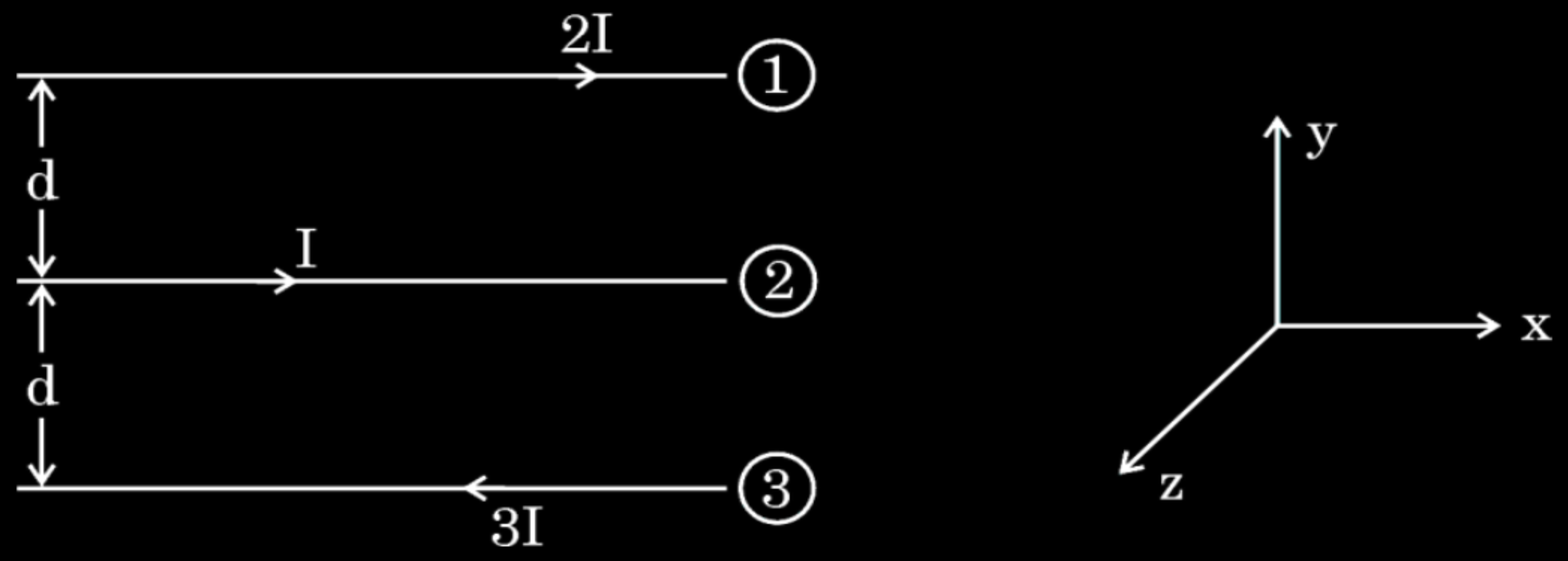
$$= \{2(50 \times 10^{-2})\,\hat{\imath} \times (-0.5)\,\hat{k}\} + \{2(20 \times 10^{-2})(-\hat{\jmath}) \times (-0.5)\,\hat{k}\}$$

$$= 0.5\hat{\jmath} + 0.2\hat{\imath}$$

$$F = \sqrt{(F_1^2 + F_2^2)} = \sqrt{(0.25 + 0.04)} = \sqrt{0.29}\text{ N}$$

23. (a) The figure given below shows three straight long parallel conductors (1), (2) and (3) kept in x-y plane, carrying currents 2I, I and 3I respectively as shown in figure.

CBSE 2026



Find the magnitude and direction of :

(i) net magnetic field at a point on conductor (1) and

(ii) net magnetic force acting on unit length of conductor (1), due to conductors (2) and (3). *3*

(i) Magnetic field due to conductor (2) at a point on conductor (1) is $B_1 = \frac{\mu_0 I}{2\pi d}$ along positive z - axis. ½

Magnetic field due to conductor (3) at a point on conductor (1) is $B_2 = \frac{3\mu_0 I}{4\pi d}$ along negative z - axis. ½

Magnitude of net magnetic field is $B_2 - B_1 = \frac{3\mu_0 I}{4\pi d} - \frac{\mu_0 I}{2\pi d} = \frac{\mu_0 I}{4\pi d}$ along negative z – axis. ½

Direction will be along negative z axis/ into the plane of the paper.

(ii) Force per unit length on conductor (1) due to conductor (2) $= \frac{\mu_0 I_1 I_2}{2\pi d}$

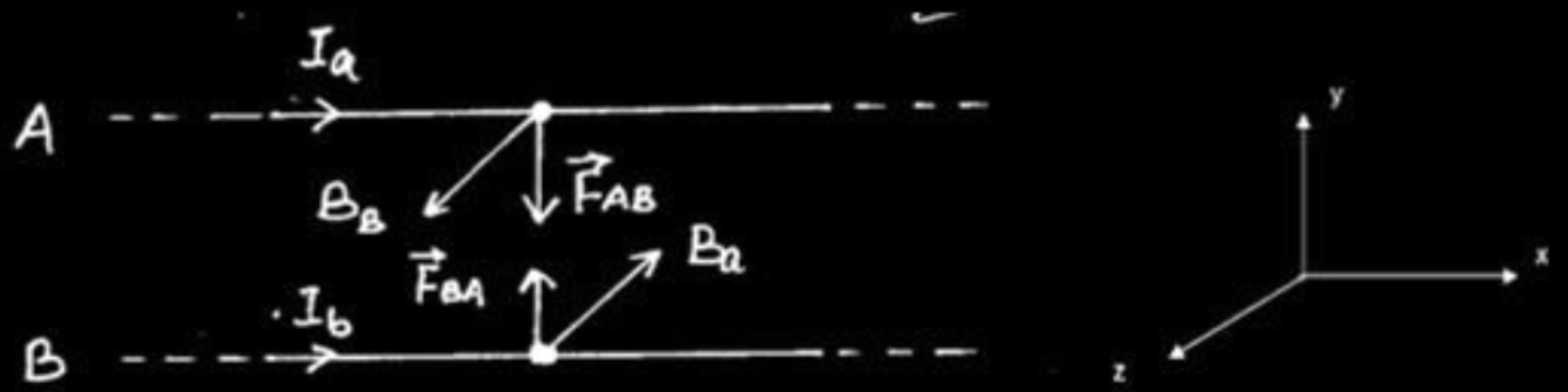
$= \frac{\mu_0 I^2}{\pi d}$ (attractive) ½

Force per unit length on conductor (1) due to conductor (3) $= \frac{3\mu_0 I^2}{2\pi d}$ (repulsive)

Magnitude of net force per unit length on conductor (1) due to (2) & (3) $= \frac{\mu_0 I^2}{2\pi d}$ ½

Direction of the net force per unit length will be repulsive or along positive y-axis. ½

Two long straight parallel conductors A and B carrying steady currents I_a and I_b in the same direction are separated by a distance d. Deduce the expressions for the force acting on length L of conductor B due to conductor A and show it in figure. Write the expression for the force acting on length L of conductor A due to conductor B and show that it follows Newton's third law. CBSE 2026



1

Note: Deduct ½ mark if direction of force on conductor B is not shown.

Magnetic field due to current carrying conductor A at all points along the conductor B

$$\overrightarrow{B_a} = \frac{\mu_0 I_a}{2\pi d}(-\hat{k})$$ ½

Force experienced by a segment of length L of conductor B

$$\vec{F}_{BA} = I_b \vec{L} \times \vec{B}_a$$

$$\vec{F}_{BA} = I_b L\left(\frac{\mu_0 I_a}{2\pi d}\right)[\hat{\imath} \times (-\hat{k})]$$

$$\vec{F}_{BA} = \frac{\mu_0 I_a I_b L}{2\pi d}\hat{j}$$ ½

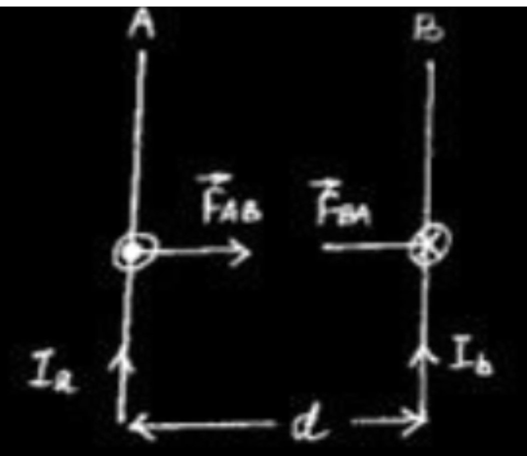
Similarly

$$\vec{F}_{AB} = \frac{\mu_0 I_a I_b L}{2\pi d}(-\hat{j})$$ ½

$$\vec{F}_{BA} = -\vec{F}_{AB}$$ ½

It follows Newton's third law

Alternatively:



$B_a = \frac{\mu_0 I_a}{2\pi d}$ Vertically inward

$F_{BA} = I_b\, L\, B_a = \frac{\mu_0 I_a I_b}{2\pi d} L$ directed towards A

$F_{AB} = I_a L\, B_b$

$F_{AB} = \frac{\mu_0 I_a I_b L}{2\pi d}$

It is equal in magnitude to F_{BA} but directed towards the conductor'B'.

Thus $\vec{F}_{BA} = -\vec{F}_{AB}$

Thus, they follow Newton's third law.

Note: Award 1 mark for the diagram, if the directions are not indicated in the diagram but written

32. (a) (i) Two long parallel straight conductors carrying current I_1 and I_2 are kept r distance apart in air. Obtain an expression for the force per unit length on one conductor due to the magnetic field produced by the other conductor. Hence, define one ampere. Under what condition will the force between the conductors be attractive in nature ?

(ii) A closely wound circular coil of radius 6·28 cm, having 50 turns carries a steady current of 4 A. Find the magnetic field at the centre of the coil. How will the magnitude of the magnetic field be affected if the radius of the coil is halved keeping other factors same ? *5*

(i) Obtaining expression of force/length	1½
Definition of one ampere	1
Condition of force to be attractive	½
(ii) Calculation of magnetic field	1½
Effect on magnitude of magnetic field when radius is halved	½

(i) Consider two long parallel conductors P and Q, distance r apart carrying currents I_1 and I_2 respectively.

magnetic field of P at a point in the position of Q

$$B_1 = \frac{\mu_o I_1}{2\pi r} \quad \text{\_\_\_\_\_\_\_\_(i)}$$

Magnetic force on Q at length l

$$F_{QP} = B_1 I_2 l = \frac{\mu_o I_1 I_2}{2\pi r} l \quad \text{\_\_\_\_\_\_\_\_(ii)} \qquad ½$$

$$f_2 = \frac{F_{QP}}{l} = \frac{\mu_o I_1 I_2}{2\pi r} \quad \text{\_\_\_\_\_\_\_\_(iii)} \qquad ½$$

Similarly force per unit length of P due to magnetic field of Q

$$f_1 = \frac{F_{PQ}}{l} = \frac{\mu_o I_1 I_2}{2\pi r} \quad \text{\_\_\_\_\_\_\_\_(iv)} \qquad ½$$

From eq. (ii) and (iv)

$$f_1 = f_2 = f = \frac{\mu_o I_1 I_2}{2\pi r} \quad \text{\_\_\_\_\_\_\_\_(v)}$$

Definition of one ampere

One ampere is the value of the steady current which when maintained in each of the two very long straight parallel conductors of negligible cross section, placed one meter apart in vaccum, would produce on each of these conductors a force equal to 2×10^{-7} newton per meter of length. 1

The force between the two conductors will be attractive when they carry current in the same direction. ½

(ii) Magnetic field at the centre of the current carrying coil

$$B = N\frac{\mu_0 I}{2a} \qquad ½$$

a= 6.28 cm N = 50 I = 4A

$$B = \frac{50\times4\pi\times10^{-7}\times4}{2\times6.28\times10^{-2}} \qquad ½$$

$$= 4\times10^{-3}\,\text{T} \qquad ½$$

When radius of the coil is halved i.e. $\frac{a}{2}$ then magnetic field becomes two times. $B = 8\times10^{-3}$ T ½

29. Two infinitely long parallel conductors A and B, separated by 2 cm, carry 6 A and 2 A currents in opposite directions respectively. Find :

(a) the net magnetic field at a point midway between A and B.

(b) the force acting per unit length on B. *3*

(a)Calculation of magnetic field due to current in A and B	½+½
Net magnetic field	1
(b) Force per unit length	1

(a) $B_1 = \dfrac{\mu_0 I_1}{2\pi r} = \dfrac{2\times10^{-7}\times6}{1\times10^{-2}}$ $=1.2\times10^{-4}$ T into the page	½
$B_2 = \dfrac{\mu_0 I_2}{2\pi r} = \dfrac{2\times10^{-7}\times2}{1\times10^{-2}}$ $=0.4\times10^{-4}$ T into the page	½
$\because$ B_1 and B_2 are in same direction. $B_{net} = B_1 + B_2$ $=1.2\times10^{-4}$ T + 0.4×10^{-4} T	
$=1.6\times10^{-4}$ T into the page	½+ ½
(b) Magnitude of magnetic force per unit length $\dfrac{F}{l} = \dfrac{\mu_0}{2\pi}\dfrac{I_1 I_2}{r}$	½
$=\dfrac{2\times10^{-7}\times6\times2}{2\times10^{-2}}$ $= 1.2\times10^{-4}$ N/m	½

(b) (i) Derive an expression for the force acting on a current carrying straight conductor kept in a magnetic field. State the rule which is used to find the direction of this force. Give the condition under which this force is (1) maximum, and (2) minimum.

(ii) Two long parallel straight wires A and B are 2·5 cm apart in air. They carry 5·0 A and 2·5 A currents respectively in opposite directions. Calculate the magnitude of the force exerted by wire A on a 10 cm length of wire B. *5*

i) Derivation of expression for force	2
Statement of Rule	½
Conditions for maximum and minimum force	½ + ½
ii) Calculation of magnitude of force	1 ½

Consider a rod of uniform cross sectional area A and length l. Let the number density of mobile charge carriers in it be n. ½

Thus the total number of mobile charge carriers in it is n l A.

For steady current I, drift velocity of electrons $\vec{v_d}$, in the presence of external magnetic field $\vec{B}$, the force on these carriers is ½

$$\vec{F} = n\,l\,Aq(\vec{v_d}\times\vec{B})$$ ½

$$= \left[\vec{j}Al\right]\times\vec{B}$$ ½

$$= I(\vec{l}\times\vec{B})$$

Where $nq\vec{v_d}$ is current density ($\vec{j}$) and $|\vec{j}A|$ is current (I)

Fleming's left hand Rule: If forefinger, middle finger and thumb are stretched in mutually perpendicular directions, such that forefinger indicates the direction of magnetic field, middle finger indicates the direction of current in the conductor, then thumb indicates the direction of force on the conductor.

Or

Right Hand Thumb Rule : If the fingers of right hand are made to rotate from $\vec{l}\,to\,\vec{B}$ through angle θ, the thumb points in the direction of force on the current carrying conductor. ½

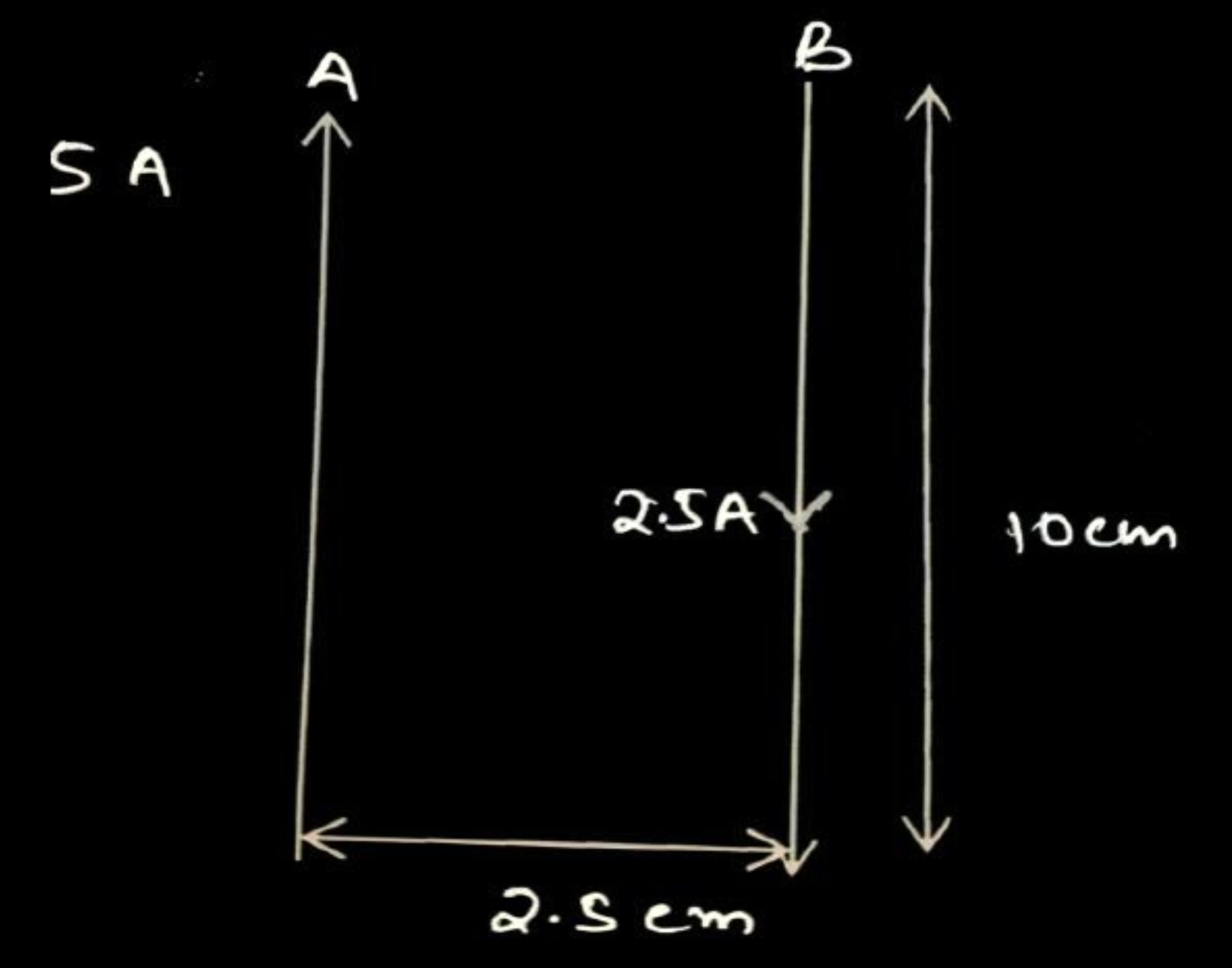
Condition for maximum force $\theta = 90^0$ ½

$$|\vec{F}| = I\,l\,B\sin\theta = I\,l\,B$$

Condition for minimum force $\theta = 0^o$ or 180^0 ½

$$|\vec{F}| = 0$$

ii)



$$F = \frac{\mu_o}{4\pi}\frac{2I_1I_2}{d}\,l$$ ½

$$= \frac{10^{-7}\times2\times5\times2.5}{2.5\times10^{-2}}\times10\times10^{-2}\,N$$ ½

$$= 10^{-5}\text{ N}$$ ½

13. Beams of electrons and protons move parallel to each other in the same direction. They **1**

(a) attract each other.

(b) repel each other.

(c) neither attract nor repel.

(d) force of attraction or repulsion depends upon speed of beams.

13. Beams of electrons and protons move parallel to each other in the same direction. They **1**

(a) attract each other.

(b) repel each other.

(c) neither attract nor repel.

(d) force of attraction or repulsion depends upon speed of beams.

(b) Repel each other

2. Two long parallel wires kept 2 m apart carry 3A current each, in the same direction. The force per unit length on one wire due to the other is **1**

(A) 4.5×10^{-5} Nm^{-1}, attractive

(B) 4.5×10^{-7} N/m, repulsive

(C) 9×10^{-7} N/m, repulsive

(D) 9×10^{-5} N/m, attractive

5. Two long straight parallel conductors carrying steady currents I_1 and I_2 in the same direction are separated by a distance d. The force exerted by one conductor on unit length of the other is :

(a) $\dfrac{\mu_0 I_1 I_2}{4\pi d}$, attractive

(b) $\dfrac{\mu_0 I_1 I_2}{4\pi d}$, repulsive

(c) $\dfrac{\mu_0 I_1 I_2}{2\pi d}$, attractive

(d) $\dfrac{\mu_0 I_1 I_2}{2\pi d}$, repulsive

5. Two long straight parallel conductors carrying steady currents I_1 and I_2 in the same direction are separated by a distance d. The force exerted by one conductor on unit length of the other is :

(a) $\dfrac{\mu_0 I_1 I_2}{4\pi d}$, attractive

(b) $\dfrac{\mu_0 I_1 I_2}{4\pi d}$, repulsive

(c) $\dfrac{\mu_0 I_1 I_2}{2\pi d}$, attractive

(d) $\dfrac{\mu_0 I_1 I_2}{2\pi d}$, repulsive

(c) $\dfrac{\mu_o I_1 I_2}{2\pi d}$, attractive

(b) (i) Derive an expression for the force acting on a conductor carrying current I in a magnetic field $\vec{B}$.

(ii) A long horizontal conductor carries a current of 30 A, in east to west direction. What are the magnitude and direction of the magnetic field due to the current 1·0 m below the conductor ? *5*

(i) Deriving expression for force on a current carrying wire	3
(ii) Finding the magnitude and direction of magnetic field	1+1

(i) Consider a wire of length l, area of cross- section A and having number density of free electrons n and carrying electric current I. Total number of charge carriers in the wire = nIA. If $\vec{v}_d$ is drift velocity of electrons, magnetic force experience by the electrons is	½
$\vec{F} = -(nIA)e\vec{v}_d \times \vec{B}$	1
As $I\vec{l} = -enIA\vec{v}_d$	½
$\vec{F} = I(\vec{l} \times \vec{B})$	1
(ii) $B = \frac{2\mu_o I}{4\pi r}$	½
$B = \frac{2\times10^{-7}\times30}{1} = 0.6\times10^{-5}T$	½
Magnetic field is directed towards **SOUTH**	1

17. Derive an expression for magnetic force $\vec{F}$ acting on a straight conductor of length L carrying current I in an external magnetic field $\vec{B}$. Is it valid when the conductor is in zig-zag form ? Justify.

Deriving an expression for magnetic force	1½
Validity and Justification for zig-zag form conductor	½

Total number of mobile charge carriers in a conductor of length L, cross-sectional area A and number density of charge carriers n:

$$= nLA$$

Force acting on the charge carriers in external magnetic field $\vec{B}$

$\vec{F} = (nAL)\, q\vec{v}_d \times \vec{B} \quad -------(1)$ ½

Where $\vec{v}_d$ is the drift velocity of the charge carriers

Current flowing

$I = v_d qnA$ ½

$I\vec{L} = \vec{v}_d qnAL \quad ------(2)$

On solving equation (1) and (2)

$\vec{F} = I(\vec{L} \times \vec{B})$ ½

Yes, because this force can be calculated by considering zig-zag conductor as a collection of linear strips $(d\vec{l})$ and summing them vectorically. ½

5. Two long straight parallel conductors A and B, kept at a distance r, carry current I in opposite directions. A third identical conductor C, kept at a distance $\left(\frac{r}{3}\right)$ from A carry current I_1 in the same direction as in A. The net magnetic force on unit length of C is **1**

(A) $\frac{3\mu_0 I\, I_1}{2\pi r}$, towards A

(B) $\frac{3\mu_0 I\, I_1}{2\pi r}$, towards B

(C) $\frac{3\mu_0 I\, I_1}{4\pi r}$, towards A

(D) $\frac{3\mu_0 I\, I_1}{4\pi r}$, towards B

5. Two long straight parallel conductors A and B, kept at a distance r, carry current I in opposite directions. A third identical conductor C, kept at a distance $\left(\frac{r}{3}\right)$ from A carry current I_1 in the same direction as in A. The net magnetic force on unit length of C is **1**

(A) $\frac{3\mu_0 I I_1}{2\pi r}$, towards A

(B) $\frac{3\mu_0 I I_1}{2\pi r}$, towards B

(C) $\frac{3\mu_0 I I_1}{4\pi r}$, towards A

(D) $\frac{3\mu_0 I I_1}{4\pi r}$, towards B

No option is correct, award 1 mark.

4. Two thin long parallel wires separated by a distance 'a' carry current 'I' in opposite directions. The wires will :

(A) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.

(B) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.

(C) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.

(D) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.

4. Two thin long parallel wires separated by a distance 'a' carry current 'I' in opposite directions. The wires will :

(A) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.

(B) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.

(C) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.

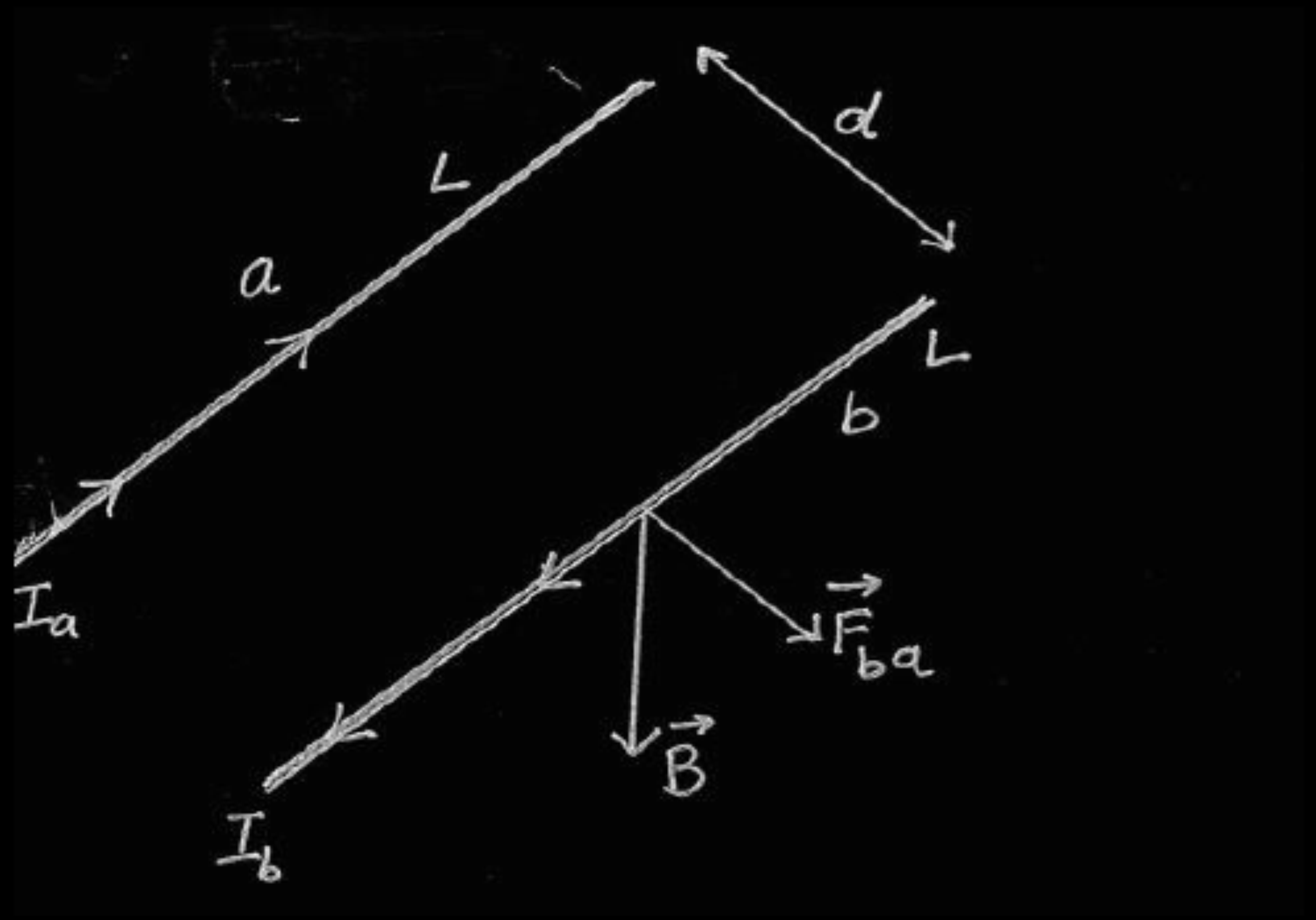
(D) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.

(D) Repel each other with a force $\frac{\mu_o I^2}{2\pi a}$, per unit length

---

23. Two long straight parallel conductors carrying currents, exert a force on each other. Why ? Derive an expression for the force per unit length between two long straight parallel conductors carrying currents in opposite directions. Explain the nature of the force between these conductors. *3*

Reason for exerting force on straight parallel conductors	½
Derivation for force per unit length	2
Explanation of nature of Force	½

One conductor experiences a force due to magnetic field of the other conductor	½
	½
Magnetic field produced by conductor ‘a’ at all points along the length of conductor ‘b’ $B_a = \frac{\mu_0 I_a}{2\pi d}$	½
Force on conductor ‘b’ due to this magnetic field $F_{ba} = I_b L B_a$	½
$F_{ba} = \frac{\mu_0 I_a I_b L}{2\pi d}$	
$f_{ba} = \frac{F_{ba}}{L} = \frac{\mu_0 I_a I_b}{2\pi d}$ directed away from a	½
$f_{ab} = \frac{F_{ab}}{L} = \frac{\mu_0 I_a I_b}{2\pi d}$ directed away from b	
Repulsive, the forces acting on them are away from each other.	½

22. (a) Two long, straight, parallel conductors carry steady currents in opposite directions. Explain the nature of the force of interaction between them. Obtain an expression for the magnitude of the force between the two conductors. Hence define one ampere. *3*

OR

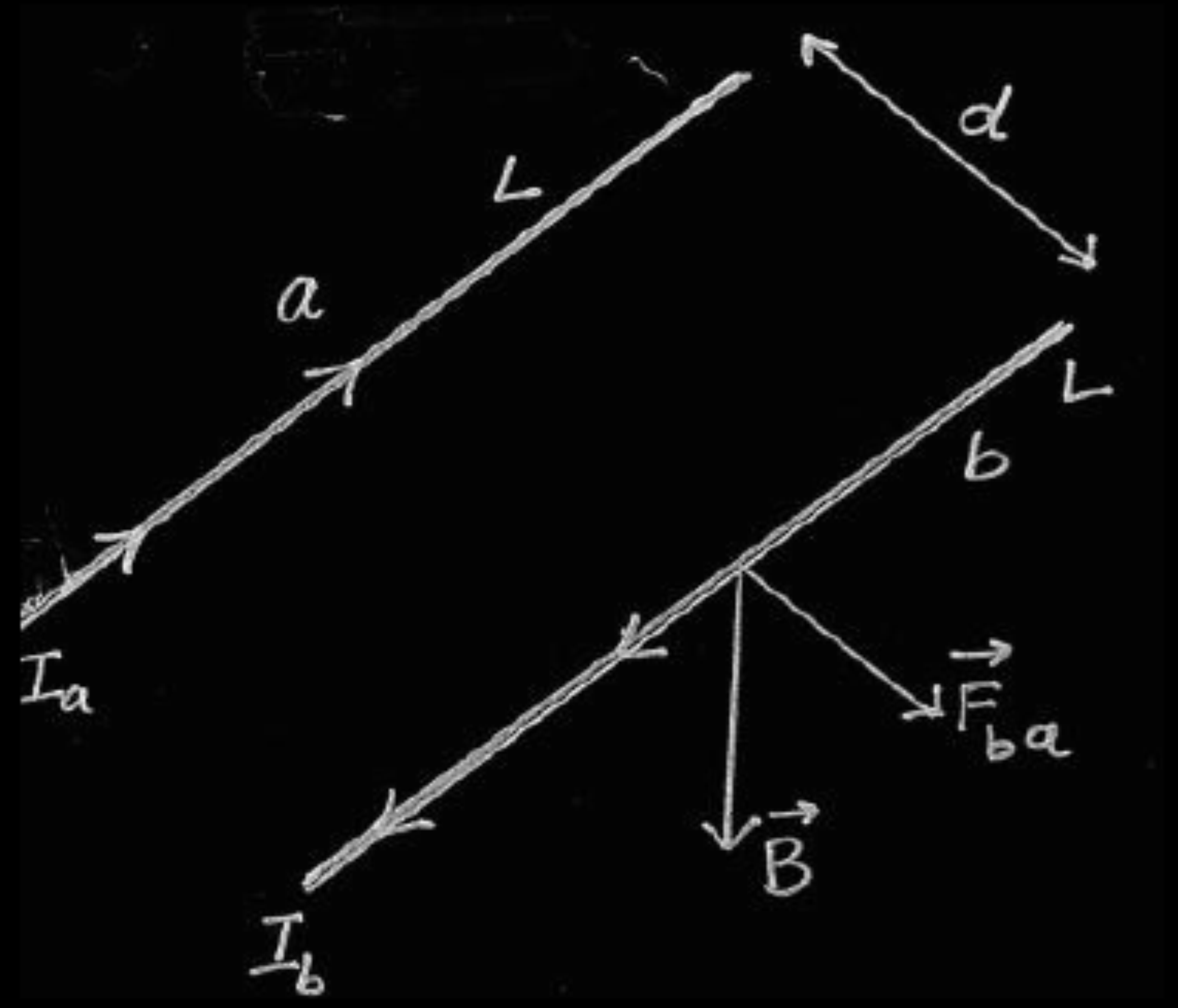
(b) Obtain an expression for the torque $\vec{\tau}$ acting on a current carrying loop in a uniform magnetic field $\vec{B}$. Draw the necessary diagram. *3*

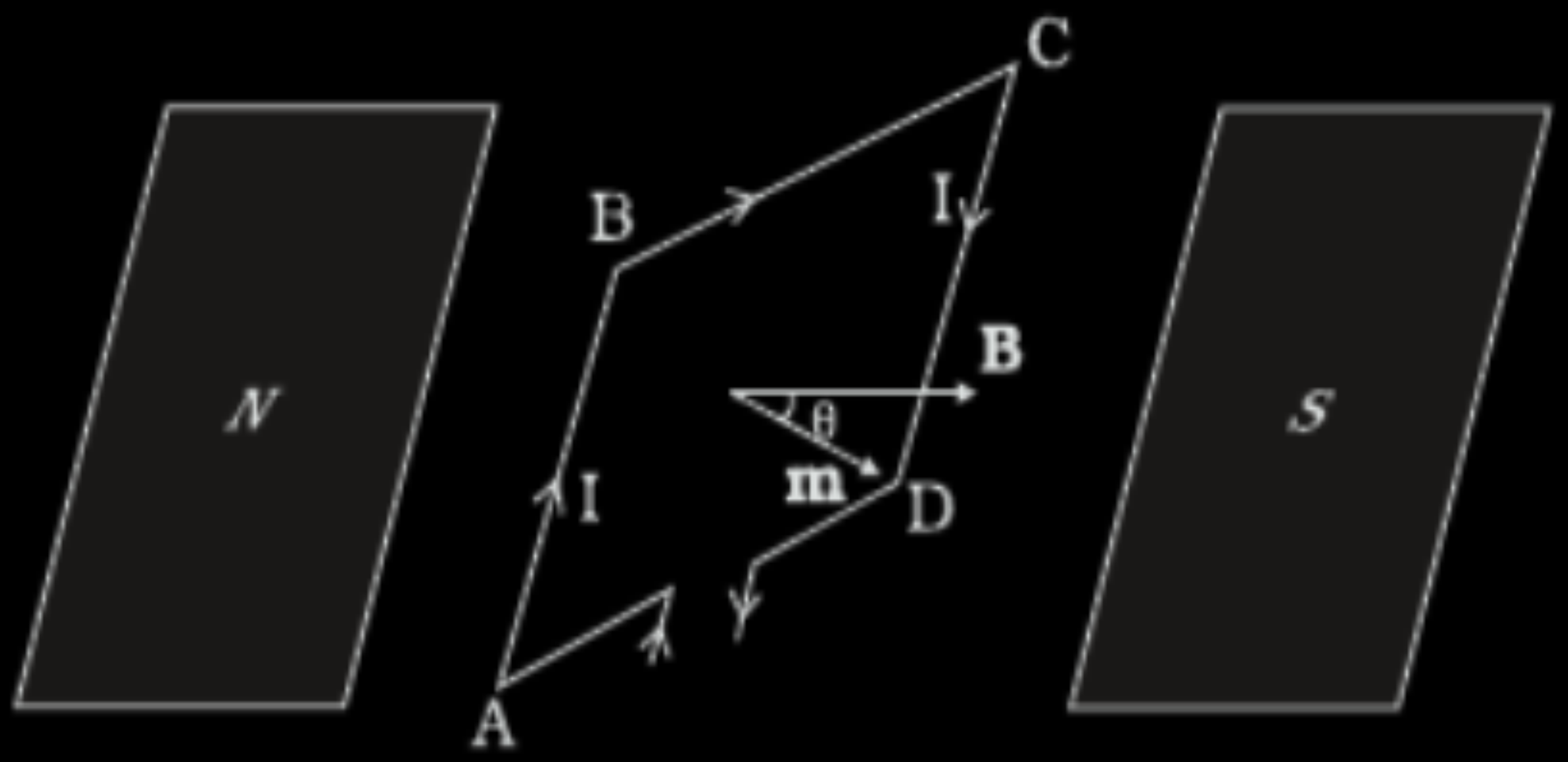
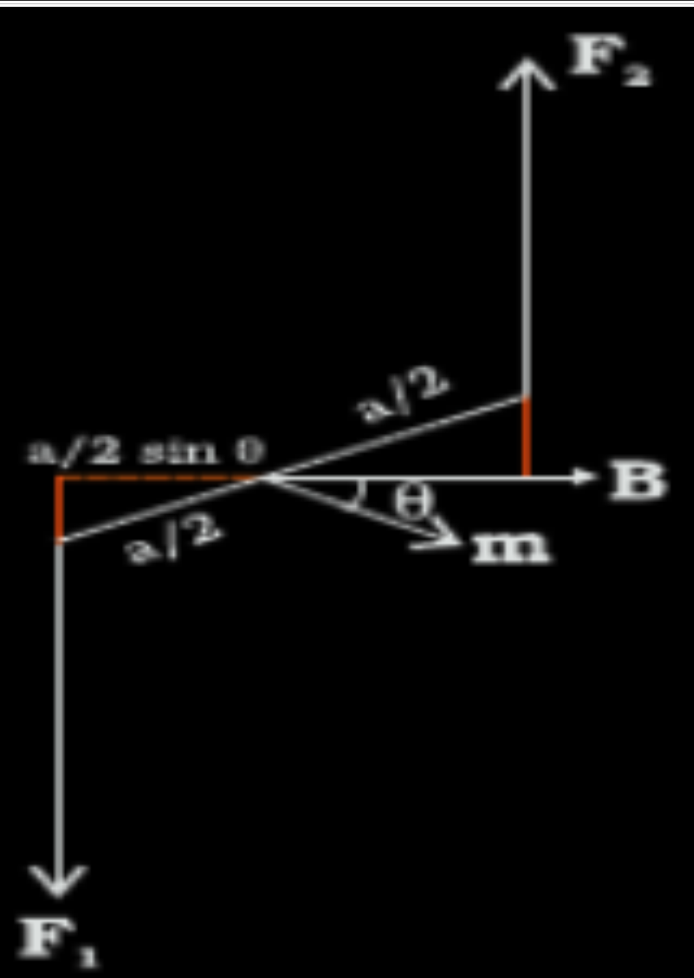
(a)

Explaining nature of force	½
Obtaining expression of force	1½
Defining one ampere	1

(b)

Obtaining expression of torque	2
Drawing diagram	1

Nature of force is repulsive.	½
	½
Magnetic field due to current I_a at all points of conductor b $B_{ab} = \frac{\mu_0 I_a}{2\pi d}$ directed downwards	½
Force experienced by conductor b on its segment of length l $F_{ab} = I_b l B_{ab}$ $= \frac{\mu_0 I_a I_b}{2\pi d} l$ directed towards left	½
Similarly Force experienced by conductor a on its segment of length l $F_{ba} = \frac{\mu_0 I_a I_b}{2\pi d} l$ directed towards right	
One ampere is that steady current which when maintained in each of two very long straight parallel conductors of negligible cross- section, placed one metre apart in vacuum produces a force of 2×10^{-7} N/m on each conductor.	1

 	1
	½
Forces on arm BC and DA are equal and opposite and act along the axis of the coil. Being collinear they cancel each other. Forces on arms AB and CD are equal and opposite but not collinear. They form a couple.	½
$F_1 = F_2 = IbB$	
$\tau = F_1 \dfrac{a}{2} \sin\theta + F_2 \dfrac{a}{2} \sin\theta$	½
$\tau = IabB\sin\theta$ $\tau = IAB\sin\theta$ (where A = ab & m = IA) $\vec{\tau} = \vec{m} \times \vec{B}$	½

6. A 10 cm long wire lies along y-axis. It carries a current of 1·0 A in positive y-direction. A magnetic field $\vec{B} = (5 \text{ mT})\hat{j} - (8 \text{ mT})\hat{k}$ exists in the region. The force on the wire is :

(A) $(0{\cdot}8 \text{ mN})\hat{i}$

(B) $-(0{\cdot}8 \text{ mN})\hat{i}$

(C) $(80 \text{ mN})\hat{i}$

(D) $-(80 \text{ mN})\hat{i}$

6. A 10 cm long wire lies along y-axis. It carries a current of 1·0 A in positive y-direction. A magnetic field $\vec{B} = (5\ \text{mT})\hat{j} - (8\ \text{mT})\hat{k}$ exists in the region. The force on the wire is :

(A) $(0{\cdot}8\ \text{mN})\hat{i}$

(B) $-(0{\cdot}8\ \text{mN})\hat{i}$

(C) $(80\ \text{mN})\hat{i}$

(D) $-(80\ \text{mN})\hat{i}$

(B) $(-0.8\ \text{mN})\hat{i}$

13. *Assertion (A) :* Two long parallel wires, freely suspended and connected in series to a battery, move apart.

Reason (R) : Two wires carrying current in opposite directions repel each other.

13. *Assertion (A) :* Two long parallel wires, freely suspended and connected in series to a battery, move apart.

Reason (R) : Two wires carrying current in opposite directions repel each other.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A)

15. *Assertion (A) :* When current is passed through a spring, two adjacent turns move towards each other.

Reason (R) : There is a force between any two adjacent turns of a current carrying spring.

15. *Assertion (A) :* When current is passed through a spring, two adjacent turns move towards each other.

Reason (R) : There is a force between any two adjacent turns of a current carrying spring.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

23. Two parallel straight wires of lengths 15 m and 20 m carrying currents 6·0 A and 8·0 A respectively in opposite directions, are 10 cm apart. Calculate the force on a 10 cm long section of the second wire near its centre. Also mention the nature of the force between these wires. *3*

Calculation of force on wire	2
Nature of the force	1

$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$	1
$= \frac{4\pi\times10^{-7}\times6\times8}{2\pi\times10\times10^{-2}}$	
$= 96\times10^{-6} N/m$	½
Force on 10 cm long section of II wire	
$F = 96\times10^{-6}\times10\times10^{-2}$	
$= 96\times10^{-7} N$	½
Nature of force is repulsive.	1

7. A current of 5 A is flowing in a wire of length 1.5 cm. A force of 7.5 mN acts on it when it is placed in a uniform magnetic field of 0.2 T. The angle between the magnetic field and the direction of current is :

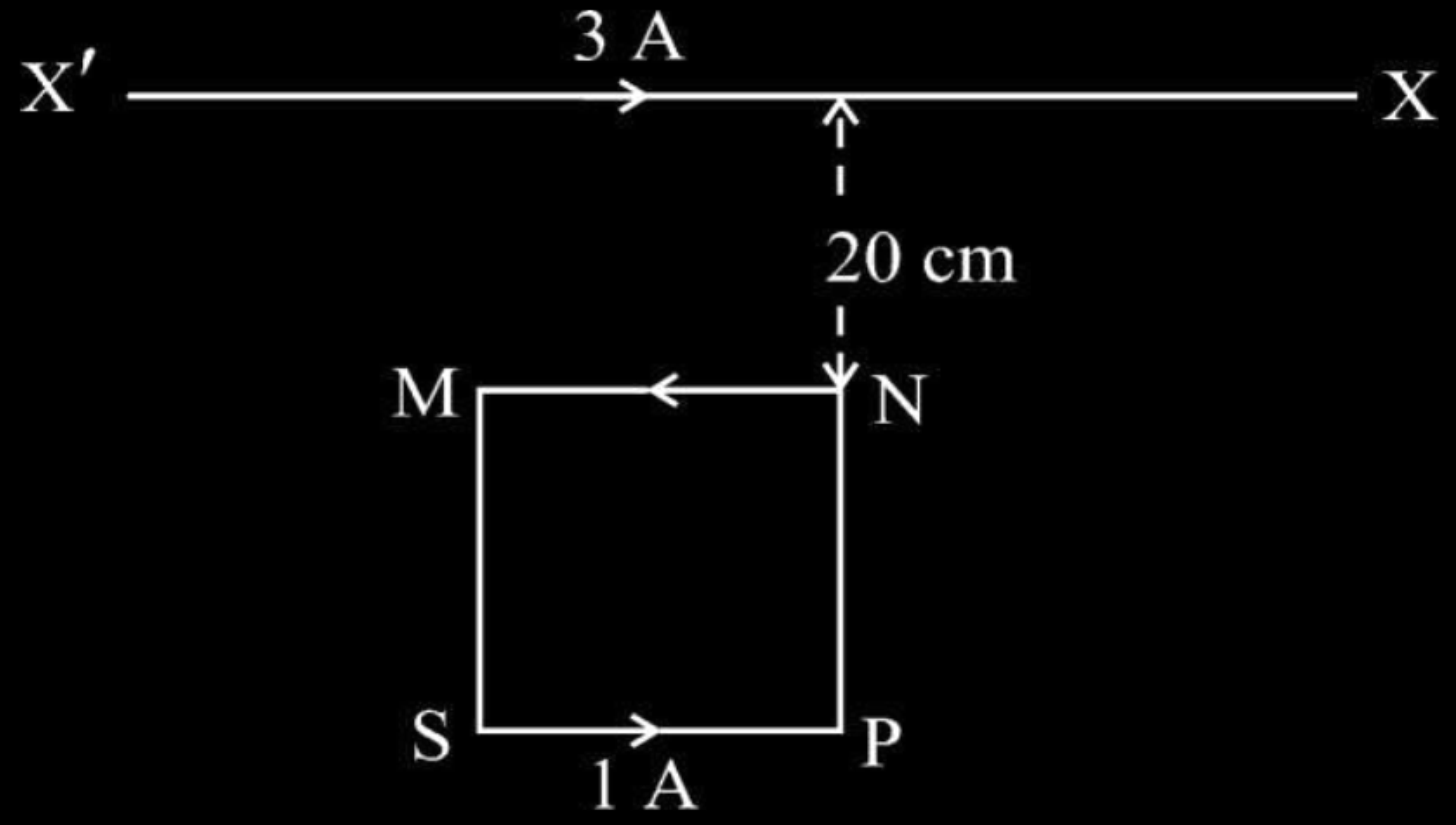
(A) 30° (B) 45° (C) 60° (D) 90°

7. A current of 5 A is flowing in a wire of length 1.5 cm. A force of 7.5 mN acts on it when it is placed in a uniform magnetic field of 0.2 T. The angle between the magnetic field and the direction of current is :

(A) 30° (B) 45° (C) 60° (D) 90°

(A) 30°

(ii) A long, straight horizontal wire $X'X$ is held stationary and carries a current of 3.0 A. A square loop MNPS of side 10 cm, carrying a current of 1.0 A is kept near the wire $X'X$ as shown in the figure. Find the magnitude and direction of the net magnetic force acting on the loop due to the wire.



7. A straight wire of length 1.0 m is placed along x-axis, in a region with magnetic field $\vec{B} = (3\hat{i} + 2\hat{j})$ T. A current of 2.0 A flows in the wire along +x direction. The magnetic force acting on the wire is :

(A) 2.0 N, along z-axis

(B) 2.0 N, along –z-axis

(C) 4.0 N, along z-axis

(D) 4.0 N, along –z-axis

7. A straight wire of length 1.0 m is placed along x-axis, in a region with magnetic field $\vec{B} = (3\hat{i} + 2\hat{j})$ T. A current of 2.0 A flows in the wire along +x direction. The magnetic force acting on the wire is :

(A) 2.0 N, along z-axis

(B) 2.0 N, along –z-axis

(C) 4.0 N, along z-axis

(D) 4.0 N, along –z-axis

(C) 4.0 N, along z-axis

5. Two parallel straight wires carrying currents I_1 and I_2 are separated by a distance x. The current in the first wire is increased from I_1 to $2I_1$ and their separation is reduced to $\frac{x}{2}$. The value of force between the wires will :

(A) become four times

(B) become two times

(C) becomes one-fourth

(D) remain the same

5. Two parallel straight wires carrying currents I_1 and I_2 are separated by a distance x. The current in the first wire is increased from I_1 to $2I_1$ and their separation is reduced to $\frac{x}{2}$. The value of force between the wires will :

(A) become four times

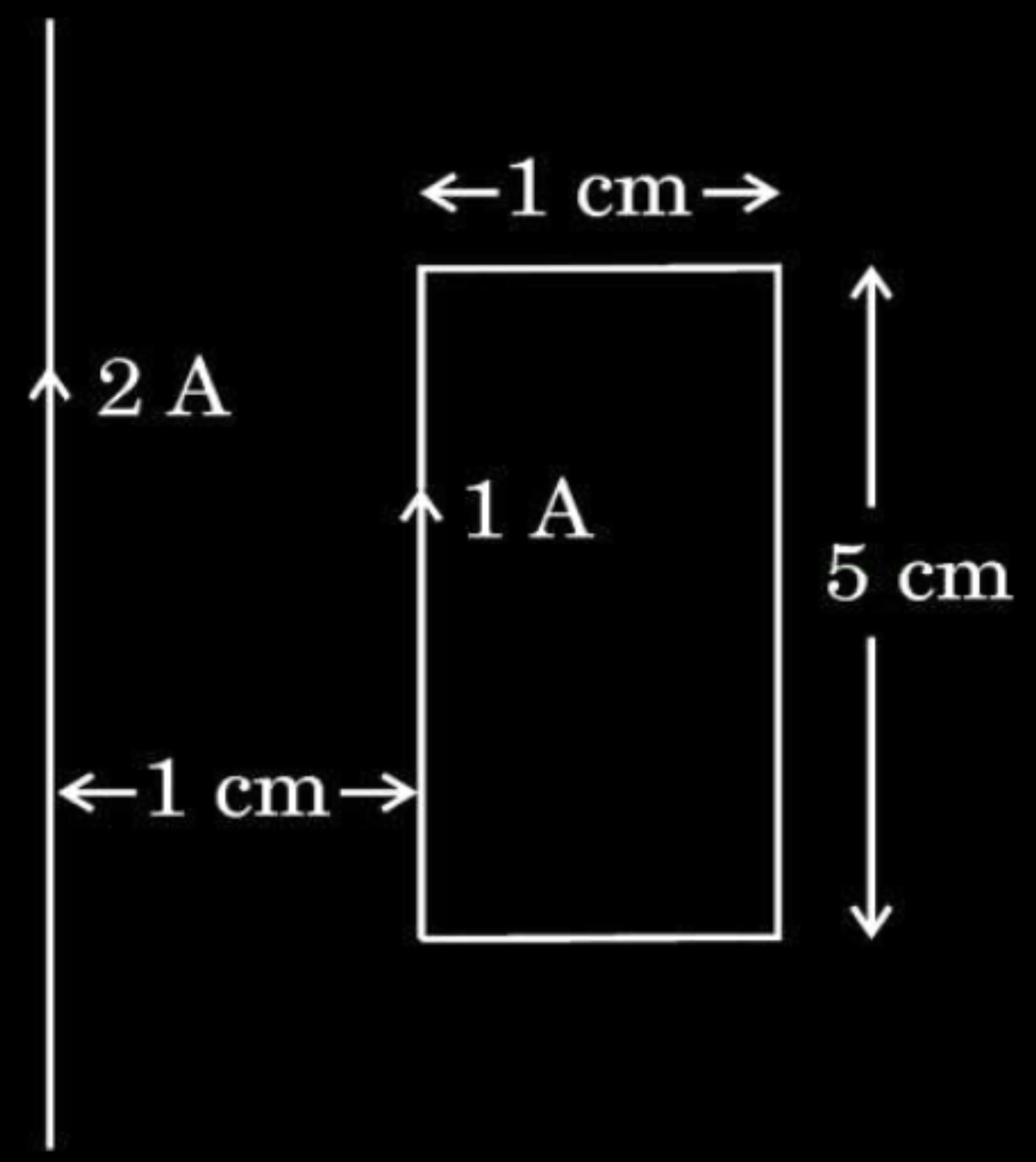
(B) become two times

(C) becomes one-fourth

(D) remain the same

(A) become four times

24. A rectangular loop carries a current of 1 A. A straight long wire carrying 2 A current is kept near the loop in the same plane as shown in the figure.



Find : *3*

(i) the torque acting on the loop, and

(ii) the magnitude and direction of the net force on the loop.

Finding-	
(i) The torque acting on the loop	1
(ii) The magnitude and direction of net force	2

(i) $\tau = mB\, sin\theta$	½
As $\vec{m}$ and $\vec{B}$ are in same direction, $\theta = 0^0$	
$\tau = 0$	½
(ii) $F = \dfrac{\mu_0 I_1 I_2\, l}{2\pi r}$	½
$F_{net} = \dfrac{\mu_0 I_1 I_2\, l}{2\pi}\left(\dfrac{1}{r_1} - \dfrac{1}{r_2}\right)$	
$= \dfrac{4\pi \times 10^{-7} \times 2 \times 1 \times 5 \times 10^{-2}}{2\pi \times 10^{-2}}\left(1 - \dfrac{1}{2}\right)$	½
$F_{net} = 1 \times 10^{-6}\, \text{N}$	½
Net force on the loop is towards the long straight wire.	½

3. A straight conductor is carrying a current of 2 A in +x direction along it. A uniform magnetic field $\vec{B} = (0{\cdot}6\,\hat{j} + 0{\cdot}8\,\hat{k})$ T is switched on, in the region. The force acting on 10 cm length of the conductor is :

(A) $(0{\cdot}12\,\hat{j} - 0{\cdot}16\,\hat{k})$ N

(B) $(-0{\cdot}16\,\hat{j} + 0{\cdot}12\,\hat{k})$ N

(C) $(-0{\cdot}12\,\hat{j} + 0{\cdot}16\,\hat{k})$ N

(D) $(0{\cdot}16\,\hat{j} - 0{\cdot}12\,\hat{k})$ N

3. A straight conductor is carrying a current of 2 A in +x direction along it. A uniform magnetic field $\vec{B} = (0{\cdot}6\,\hat{j} + 0{\cdot}8\,\hat{k})$ T is switched on, in the region. The force acting on 10 cm length of the conductor is :

(A) $(0{\cdot}12\,\hat{j} - 0{\cdot}16\,\hat{k})$ N

(B) $(-0{\cdot}16\,\hat{j} + 0{\cdot}12\,\hat{k})$ N

(C) $(-0{\cdot}12\,\hat{j} + 0{\cdot}16\,\hat{k})$ N

(D) $(0{\cdot}16\,\hat{j} - 0{\cdot}12\,\hat{k})$ N

(B) $(-0.16\hat{j}+0.12\hat{k})$N

4. A 5.0 cm long wire carrying 1.0 A current is kept in a region in which a uniform magnetic field of 0.3 T is applied perpendicular to the length of the wire. The magnitude of the force acting on the wire is – **1**

(A) zero (B) 0.45 N

(C) 0.03 N (D) 0.015 N

4. A 5.0 cm long wire carrying 1.0 A current is kept in a region in which a uniform magnetic field of 0.3 T is applied perpendicular to the length of the wire. The magnitude of the force acting on the wire is – **1**

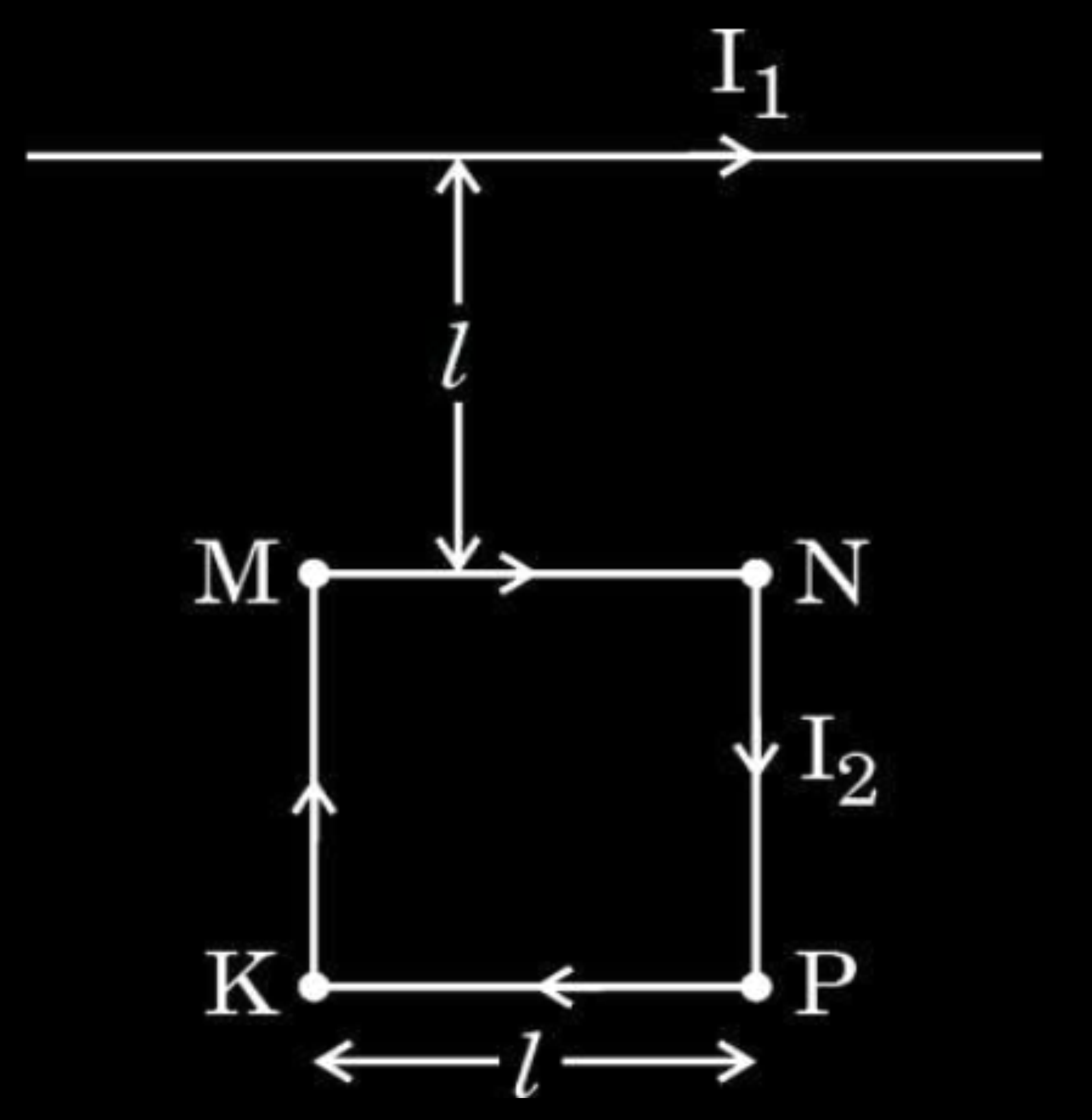
(A) zero (B) 0.45 N

(C) 0.03 N (D) 0.015 N

(D) 0.015 N

23. (a) In which cases does a charged particle not experience a force in a magnetic field ?

(b) A square loop MNPK of side 'l' carrying a current 'I_2' is kept close to a long straight wire in the same plane and the wire carries a steady current I_1 as shown in the figure. Obtain the magnitude of magnetic force exerted by the wire on the loop. *3*



(a) Conditions for, no force experienced by charged particle in magnetic field	1
(b) Obtaining the magnitude of magnetic force exerted by wire on the loop	2

(a)	
• If charged particle is at rest v = 0	½
• If B is parallel or antiparallel to v	½
(b) Magnitude of force acting between two current carrying conductor	
$F=\frac{\mu_0}{4\pi}\frac{2I_1I_2}{r}L$	½
Force on segment MN of loop	
$F_{MN}=\frac{\mu_0}{4\pi}\frac{2I_1I_2}{l}\times l$ (Towards the wire)	½
Force on segment KP of loop	
$F_{KP}=\frac{\mu_0}{4\pi}\frac{2I_1I_2}{2l}\times l$ (Away from the wire)	½
No force due to segment MK and NP	
Net force on the loop	
$F = F_{MN} - F_{KP}$	
$F = \frac{\mu_0}{4\pi}I_1I_2$	½

3. A 1 cm segment of a wire lying along x-axis carries current of 0.5 A along +x direction. A magnetic field $\vec{B} = (0.4\ mT)\ \hat{j} + (0.6\ mT)\ \hat{k}$ is switched on, in the region. The force acting on the segment is **1**

(A) $(2\hat{j} + 3\hat{k})$ mN

(B) $(-3\hat{j} + 2\hat{k})$ μN

(C) $(6\hat{j} + 4\hat{k})$ mN

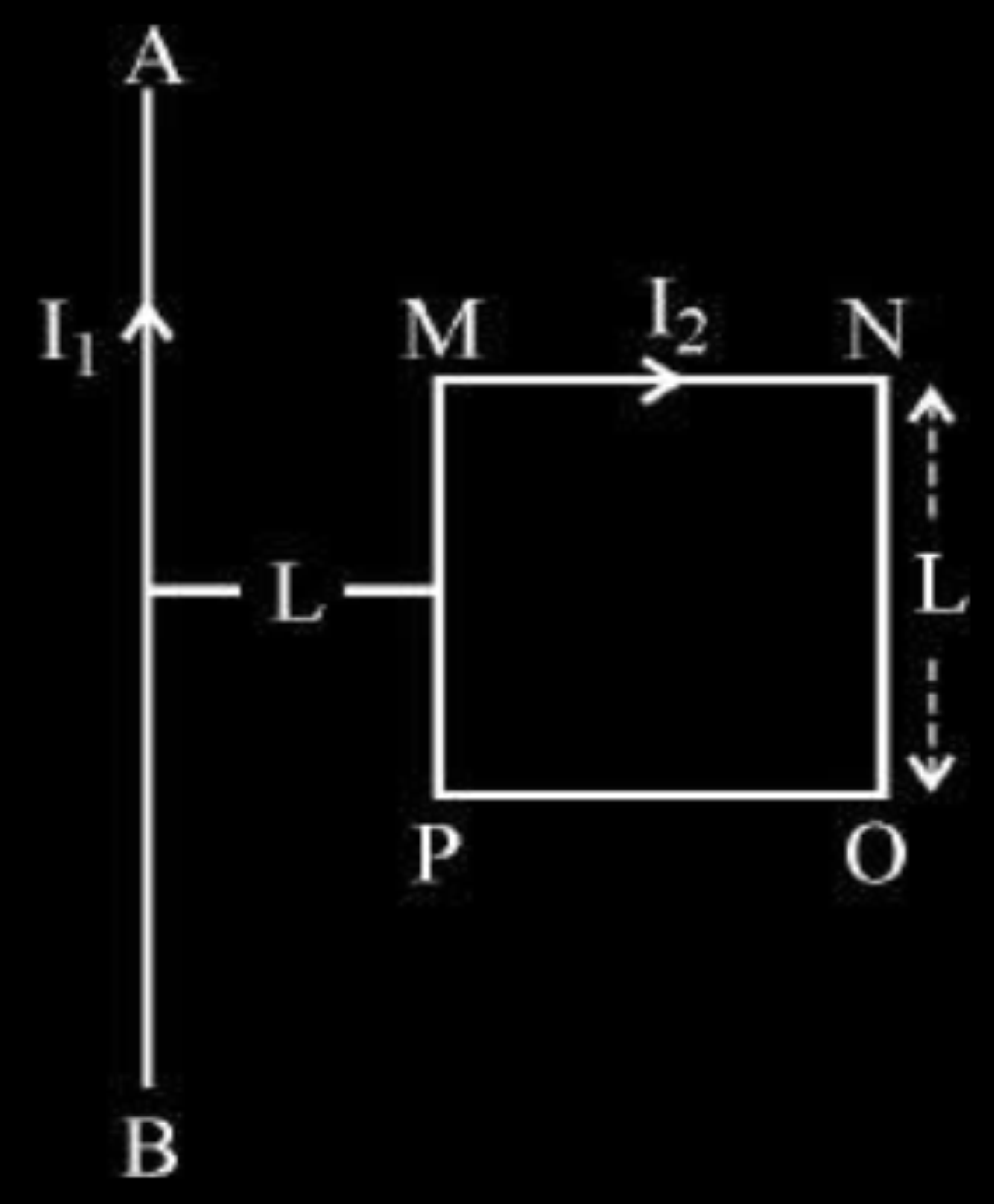
(D) $(-4\hat{j} + 6\hat{k})$ μN

3. A 1 cm segment of a wire lying along x-axis carries current of 0.5 A along +x direction. A magnetic field $\vec{B} = (0.4 \text{ mT})\,\hat{j} + (0.6 \text{ mT})\,\hat{k}$ is switched on, in the region. The force acting on the segment is 1

(A) $(2\hat{j} + 3\hat{k})$ mN
(B) $(-3\hat{j} + 2\hat{k})$ μN
(C) $(6\hat{j} + 4\hat{k})$ mN
(D) $(-4\hat{j} + 6\hat{k})$ μN

(B) $\left(-3\hat{j}+2\hat{k}\right)$μN

A square shaped current carrying loop MNOP is placed near a straight long current carrying wire AB as shown in the fig. The wire and the loop lie in the same plane. If the loop experiences a net force F towards the wire, find the magnitude of the force on the side 'NO' of the loop.

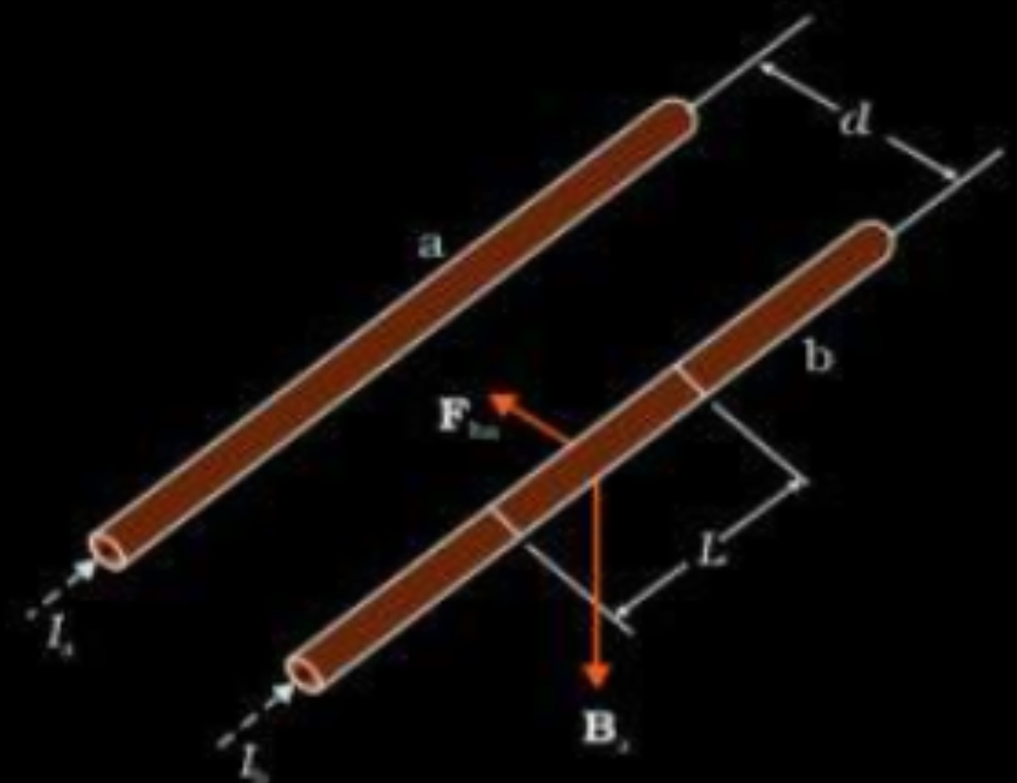


1

Magnitude of force on side NO is $= F$ **Alternatively** Let force on side MP be $= F_1$ Force on side $NO = \dfrac{F_1}{2}$	½
Magnitude of net force $= F_1 - \dfrac{F_1}{2} = \dfrac{F_1}{2} = F$ Therefore force on side NO $= \dfrac{F_1}{2} = F$	½
Give full credit if a student calculates the force as shown below. $F = \dfrac{\mu_0}{2\pi} I_1 I_2$	

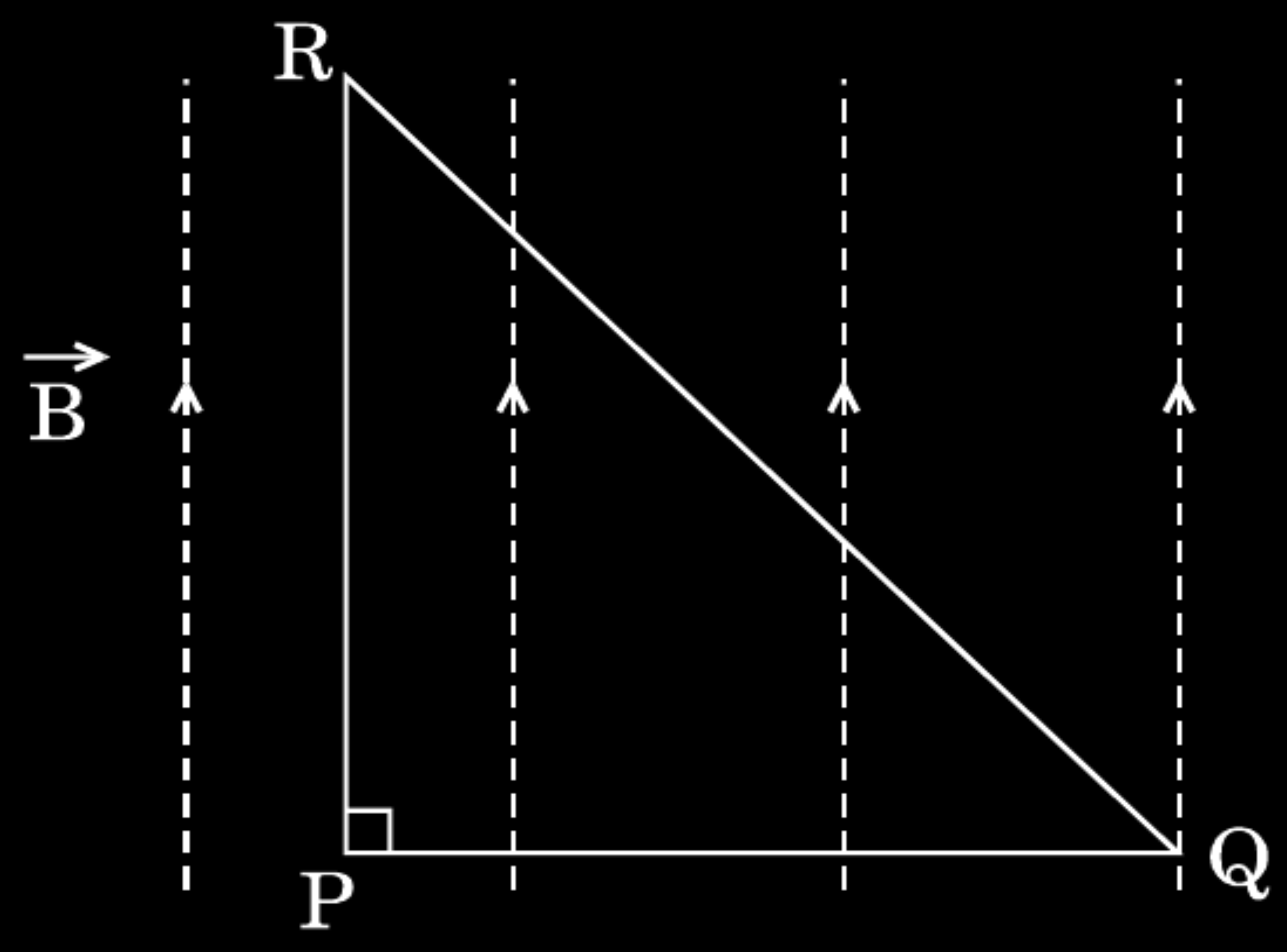
(a) Derive the expression for the force acting between two long parallel current carrying conductors. Hence, define 1 A current.

(b) A steady current of 2A flows through a circular coil having 5 turns of radius 7 cm. The coil lies in X-Y plane with its centre at the origin. Find the magnitude and direction of the magnetic dipole moment of the coil. **5**

(a)  Magnetic field due to the current I_a flowing in conductor ‘a’ at any point on conductor ‘b’ $B_a = \frac{\mu_0 I_a}{2\pi d}$ (Acting perpendicular downward) Therefore force on conductor ‘b’ due to field B_a $\vec{F} = I_b (\vec{l_b} \times \vec{B_a})$ $\lvert\vec{F_{ba}}\rvert = I_b l_b \times \frac{\mu_0 I_a}{2\pi d}$ $= \frac{\mu_0 I_a I_b l_b}{2\pi d}$ $\frac{\lvert\vec{F_{ba}}\rvert}{l_b} = \frac{\mu_0 I_a I_b}{2\pi d}$ Definition of 1 A : Two straight infinitely long parallel conductors are said to carry 1 A current each when they interact each other with a force of $2 \times 10^{-7}\ Nm^{-1}$, when kept 1m apart in vacuum	½ ½ ½ ½ 1

(b) $M = NI\pi a^2$ $= 5 \times 2 \times \frac{22}{7} \times 49 \times 10^{-4}$ $= 154 \times 10^{-3}$ $= 0.154\ Am^2$ $\vec{M}$ *will be perpendicular to* $x - y$ *plane or parallel to* Z *axis*	½ ½ ½ ½

10. An isosceles right angled current carrying loop PQR is placed in a uniform magnetic field $\vec{B}$ pointing along PR. If the magnetic force acting on the arm PQ is F, then the magnetic force which acts on the arm QR will be *1*



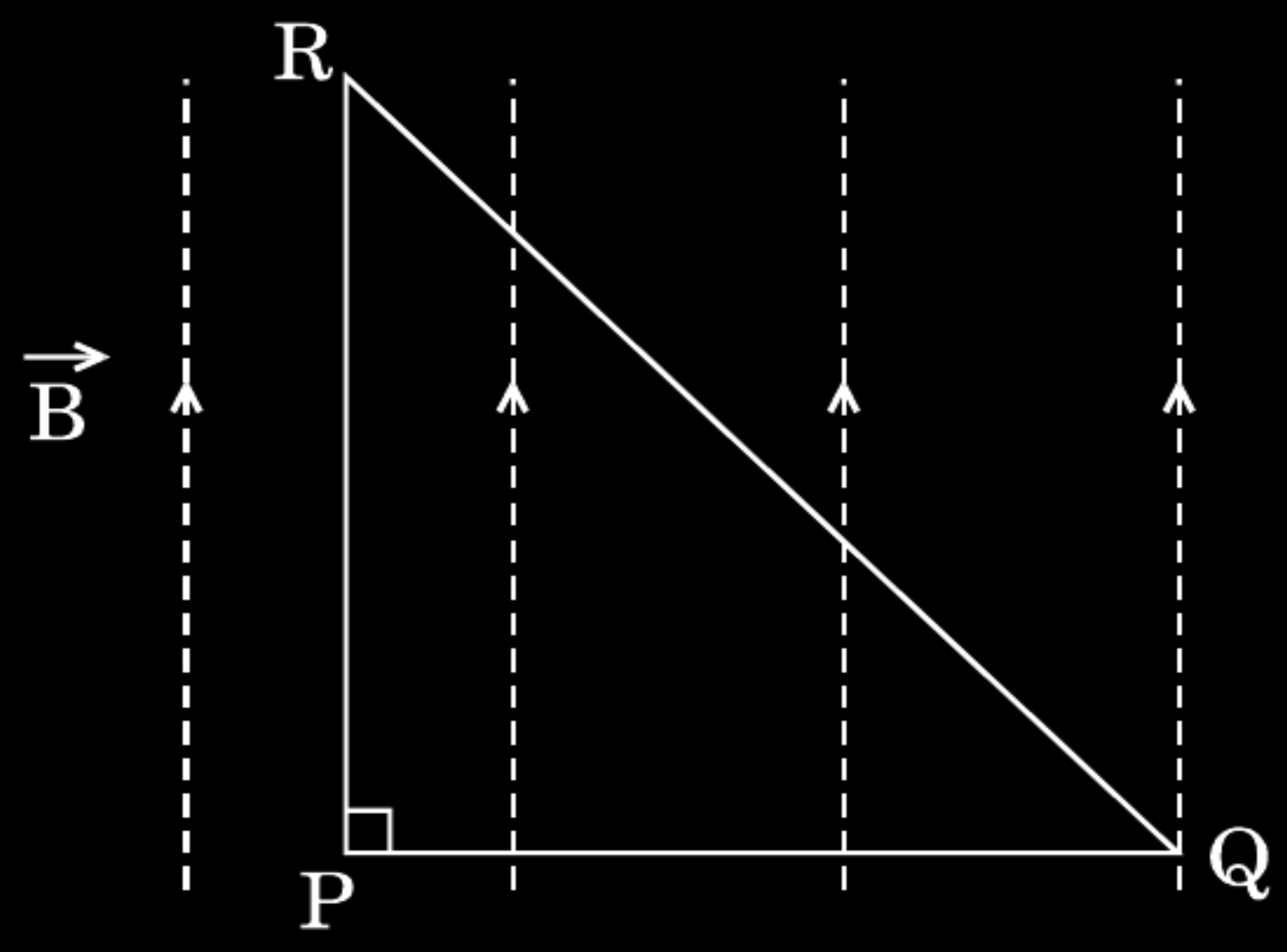
(A) F

(B) $\frac{F}{\sqrt{2}}$

(C) $\sqrt{2}\,F$

(D) $-F$

10. An isosceles right angled current carrying loop PQR is placed in a uniform magnetic field $\vec{B}$ pointing along PR. If the magnetic force acting on the arm PQ is F, then the magnetic force which acts on the arm QR will be *1*



(A) F

(B) $\frac{F}{\sqrt{2}}$

(C) $\sqrt{2}\,F$

(D) $-F$

(D)

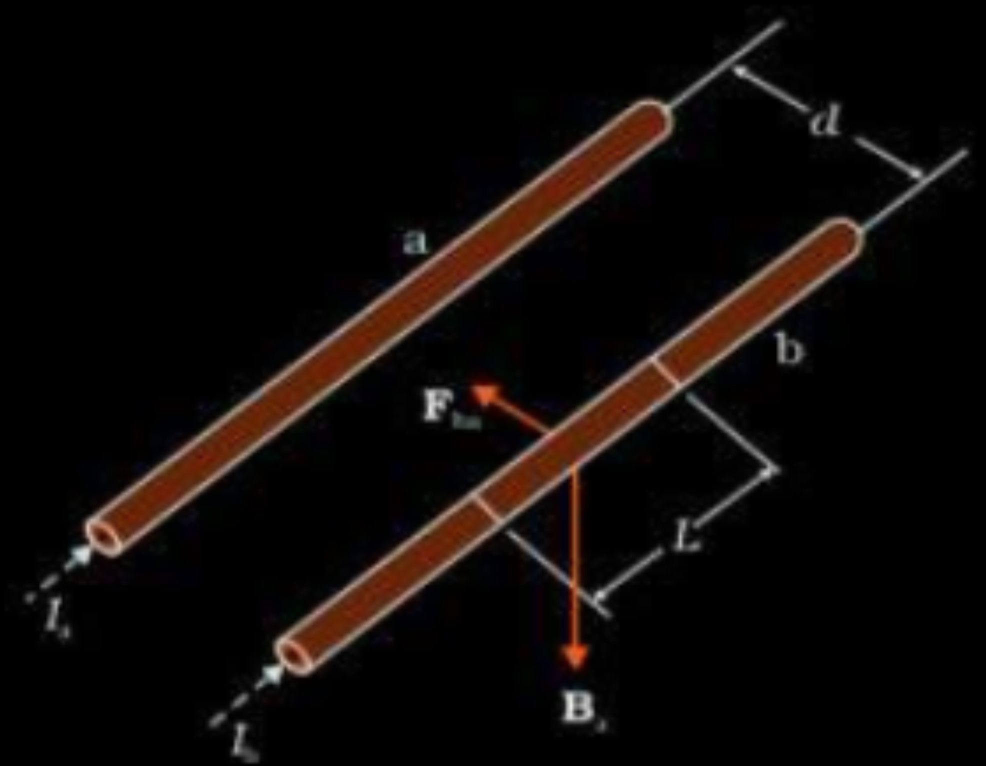
-F

36. (a) Derive the expression for the force acting per unit length between two long straight parallel current carrying conductors. Hence define one ampere.

(b) Two long parallel straight conductors are placed 12 cm apart in air. They carry equal currents of 3 A each. Find the magnitude and direction of the magnetic field at a point midway between them (drawing a figure) when the currents in them flow in opposite directions. *5*

a) Diagram showing direction	½ mark
Derivation of expression	1 ½ marks
Definition of 1 ampere	1 mark
b) Magnetic fields of the wires	½ mark
Net magnetic field and its direction	1 + ½ mark

(a)



½

Magnetic field due to the current I_a flowing in conductor 'a' at any point on conductor 'b'

$$B_a = \frac{\mu_0 I_a}{2\pi d}$$ ½

(Acting perpendicular downward)
Therefore force on conductor 'b' due to field B_a

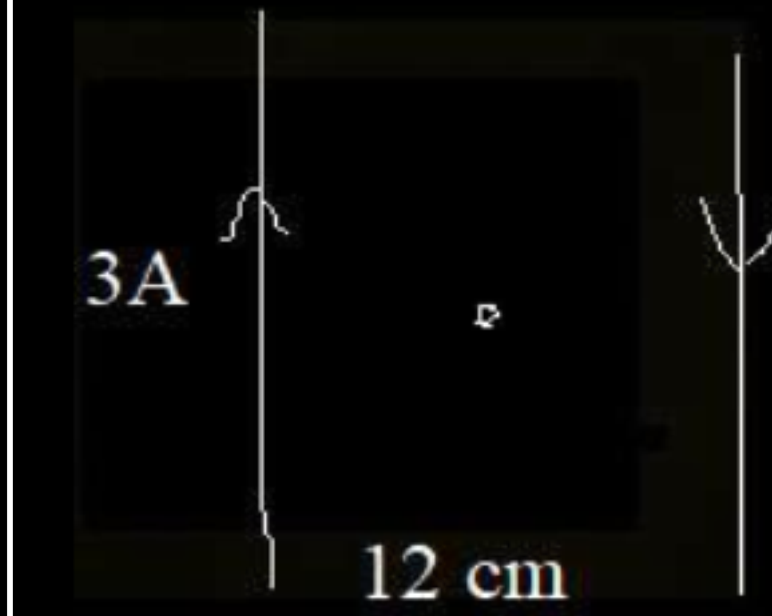
$$\vec{F} = I_b\,(\vec{l_b} \times \vec{B_a})$$ ½

$$|\overrightarrow{F_{ba}}| = I_b l_b \times \frac{\mu_0 I_a}{2\pi d}$$

$$= \frac{\mu_0 I_a I_b l_b}{2\pi d}$$

$$\frac{|\overrightarrow{F_{ba}}|}{l_b} = \frac{\mu_0 I_a I_b}{2\pi d}$$ ½

Definition of 1 A :
Two straight infinitely long parallel conductors are said to carry 1 A current each when they interact each other with a force of $2 \times 10^{-7}\ Nm^{-1}$, when kept 1m apart in vacuum 1



$$\vec{B} = \overrightarrow{B_1} + \overrightarrow{B_2}$$ ½

$$B = \frac{\mu_o I_1}{2\pi r_1} + \frac{\mu_o I_2}{2\pi r_2}$$

$$= \frac{\mu_o}{2\pi}\left(\frac{3}{6 \times 10^{-2}} + \frac{3}{6 \times 10^{-2}}\right)$$

$$= \frac{4\pi \times 10^{-7} \times 3}{\pi \times 6 \times 10^{-2}}$$

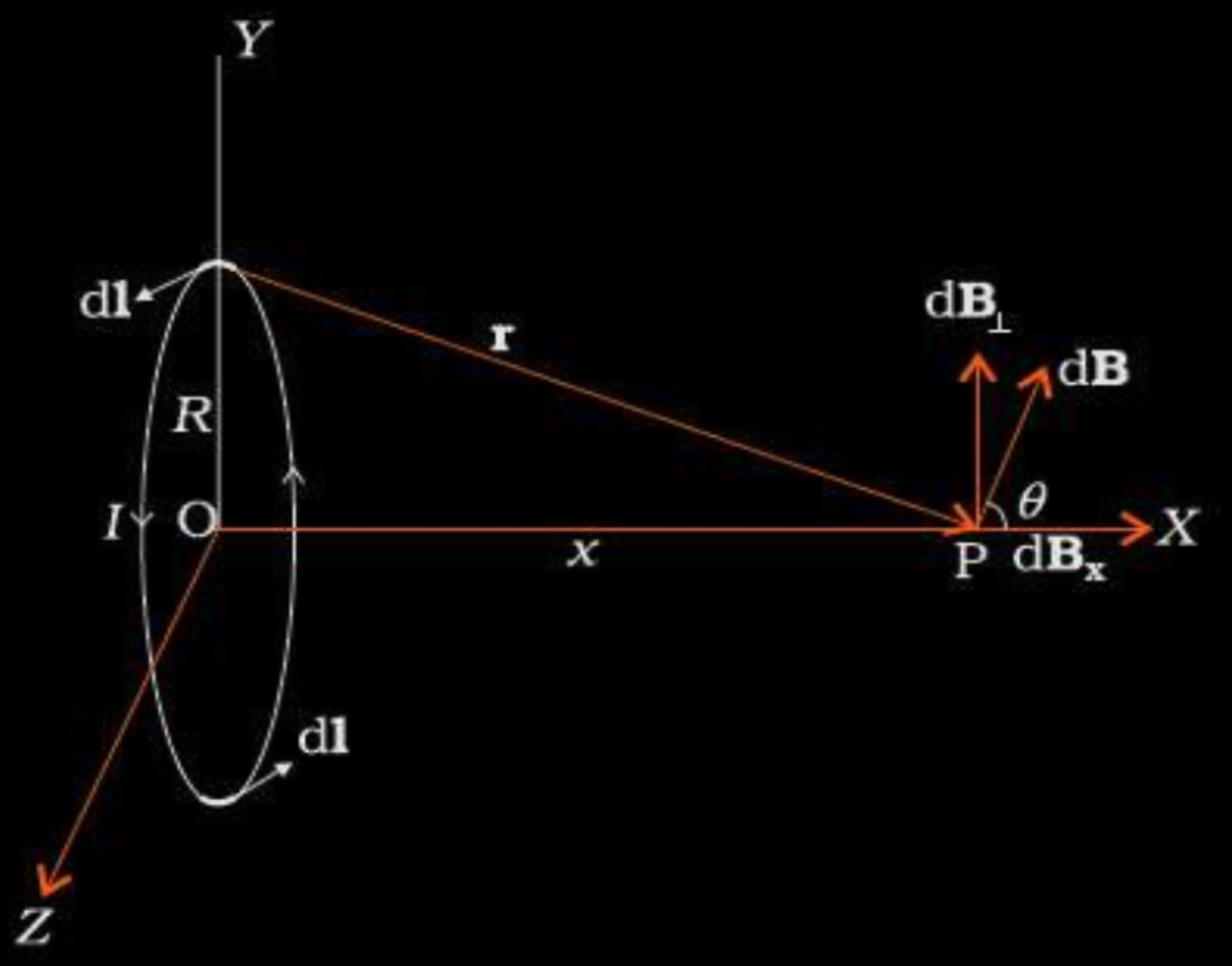
$$= 2 \times 10^{-5}\ tesla$$ 1

Direction of $\vec{B}$ at midpoint is perpendicular to the plane containing the two conductors and pointing downwards. ½
(Note: give full credit of this direction if student takes direction opposite to the shown in fig and answer accordingly)

36. (a) A circular loop of radius R carries a current I. Obtain an expression for the magnetic field at a point on its axis at a distance x from its centre.

(b) A conducting rod of length 2 m is placed on a horizontal table in north-south direction. It carries a current of 5 A from south to north. Find the direction and magnitude of the magnetic force acting on the rod. Given that the Earth's magnetic field at the place is $0{\cdot}6 \times 10^{-4}$ T and angle of dip is $\frac{\pi}{6}$. *5*

(a) Magnetic field at a point on the axis of the current loop	3
(b)Magnitude and direction of the magnetic force	2



½

$$dB = \frac{\mu_0\, Idl}{4\pi(x^2 + R^2)}$$ ½

dB has two components dB_x *and* $dB_\perp$, perpendicular components from diametrically opposite elements dl cancel out, thus only dB_x components remain effective

$$dB_x = dB\cos\theta$$ ½

$$\text{and } \cos\theta = \frac{R}{(x^2 + R^2)^{\frac{1}{2}}}$$

$$\therefore B = \int dB_x$$

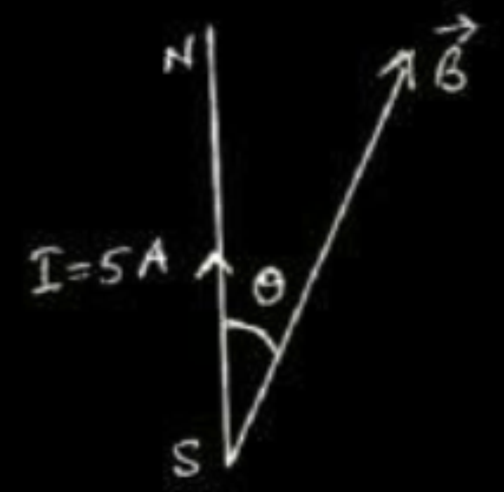
$$= \int_0^{2\pi R} \frac{\mu_0 IdlR}{4\pi(x^2 + R^2)^{\frac{3}{2}}}$$ ½

$$= \frac{\mu_0 IR^2}{2(x^2 + R^2)^{\frac{3}{2}}}$$ 1

along the axis of the loop

(b)

(i)



½

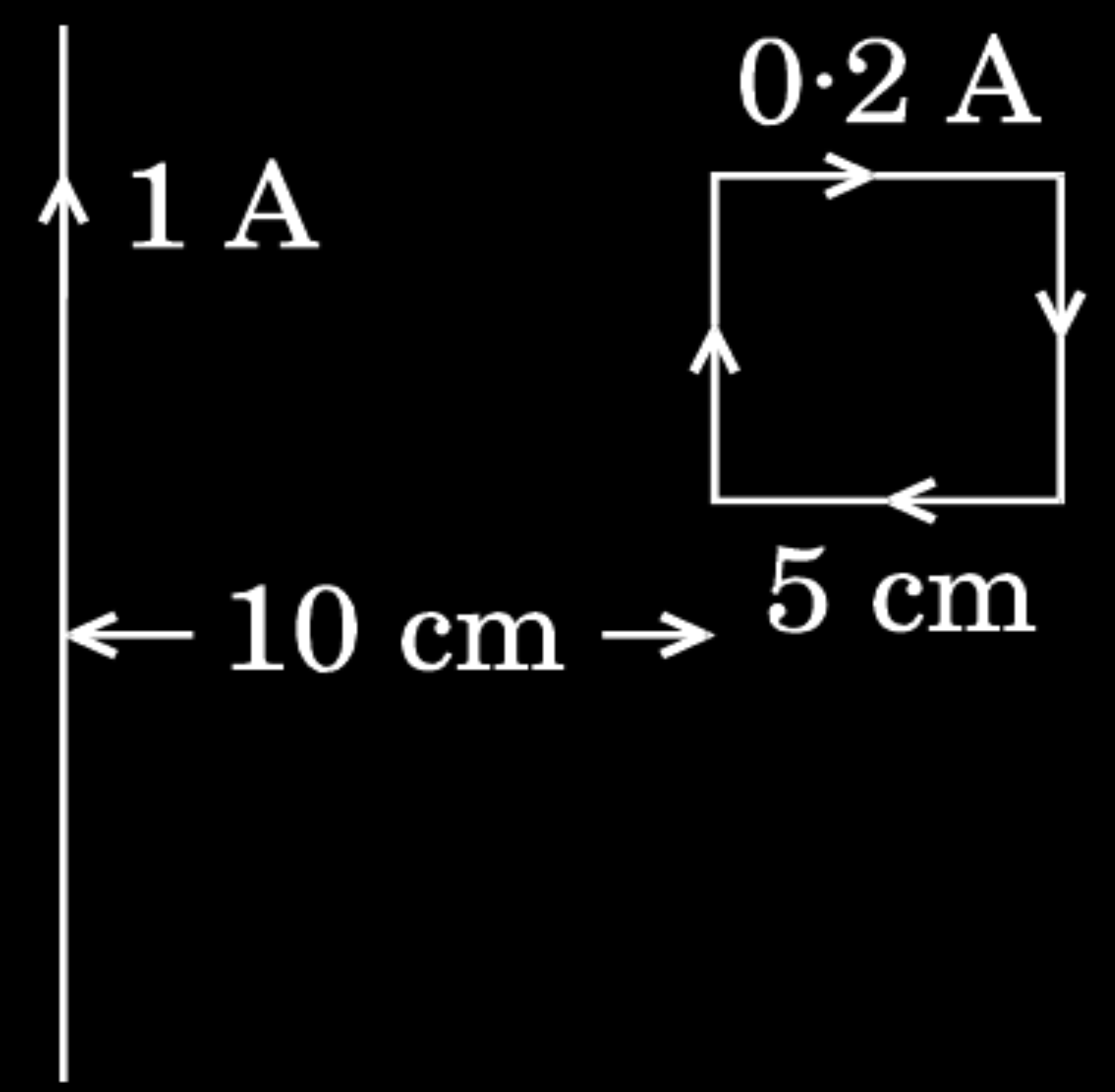
$\vec{F} = I[\vec{l} \times \vec{B}]$ ½

$F = IlB\sin\theta$

$= 5 \times 2 \times 0.6 \times 10^{-4} \times 0.5 = 3 \times 10^{-4}\text{N}$ ½

(ii) Towards east ½

(b) A square loop of sides 5 cm carrying a current of 0·2 A in the clockwise direction is placed at a distance of 10 cm from an infinitely long wire carrying a current of 1 A as shown. Calculate (i) the resultant magnetic force, and (ii) the torque, if any, acting on the loop.



Force of attraction experienced by the length SP of the loop per unit length

$$F_1 = \frac{2\mu_0 I_1 I_2}{4\pi r_1}$$ ½

$$f_1 = \frac{2\times10^{-7}\times1\times0.2}{10\times10^{-2}} = 4\times10^{-7}\,Nm^{-1}$$ ½

Force is attractive

$$f_2 = \frac{2\mu_0 I_1 I_2}{4\pi r_2}$$ ½

$$f_2 = \frac{2\times10^{-7}\times1\times0.2}{15\times10^{-2}} = 2.6\times10^{-7}\,Nm^{-1}$$ ½

Force is repulsive

So the net force experienced by the loop is (per unit length)

$$f = (f_1 - f_2)$$

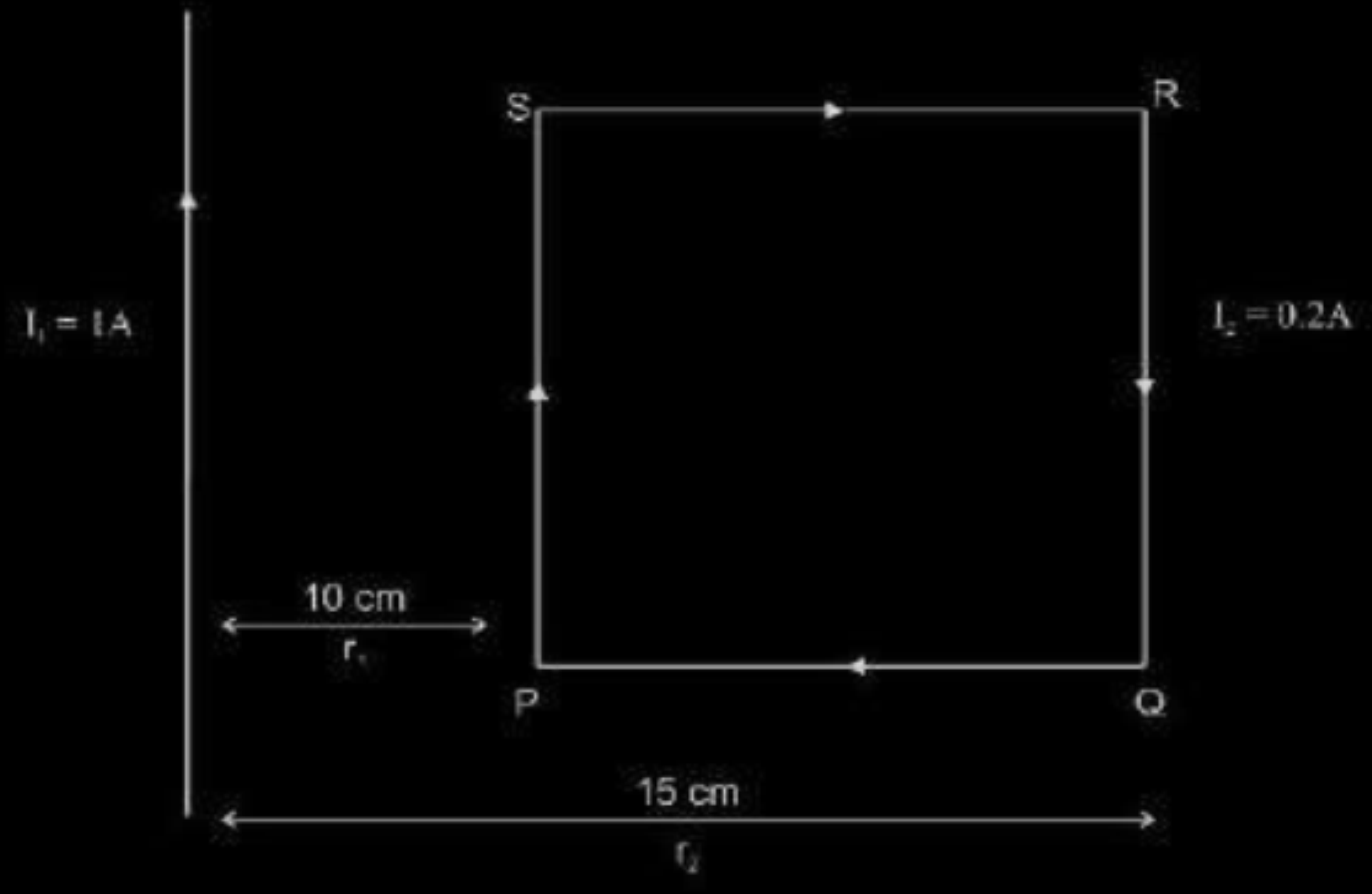
Total force experienced by the loop is:

$$F = (f_1 - f_2)l = \left(1.4\times10^{-7}\right)\times5\times10^{-2}$$ ½

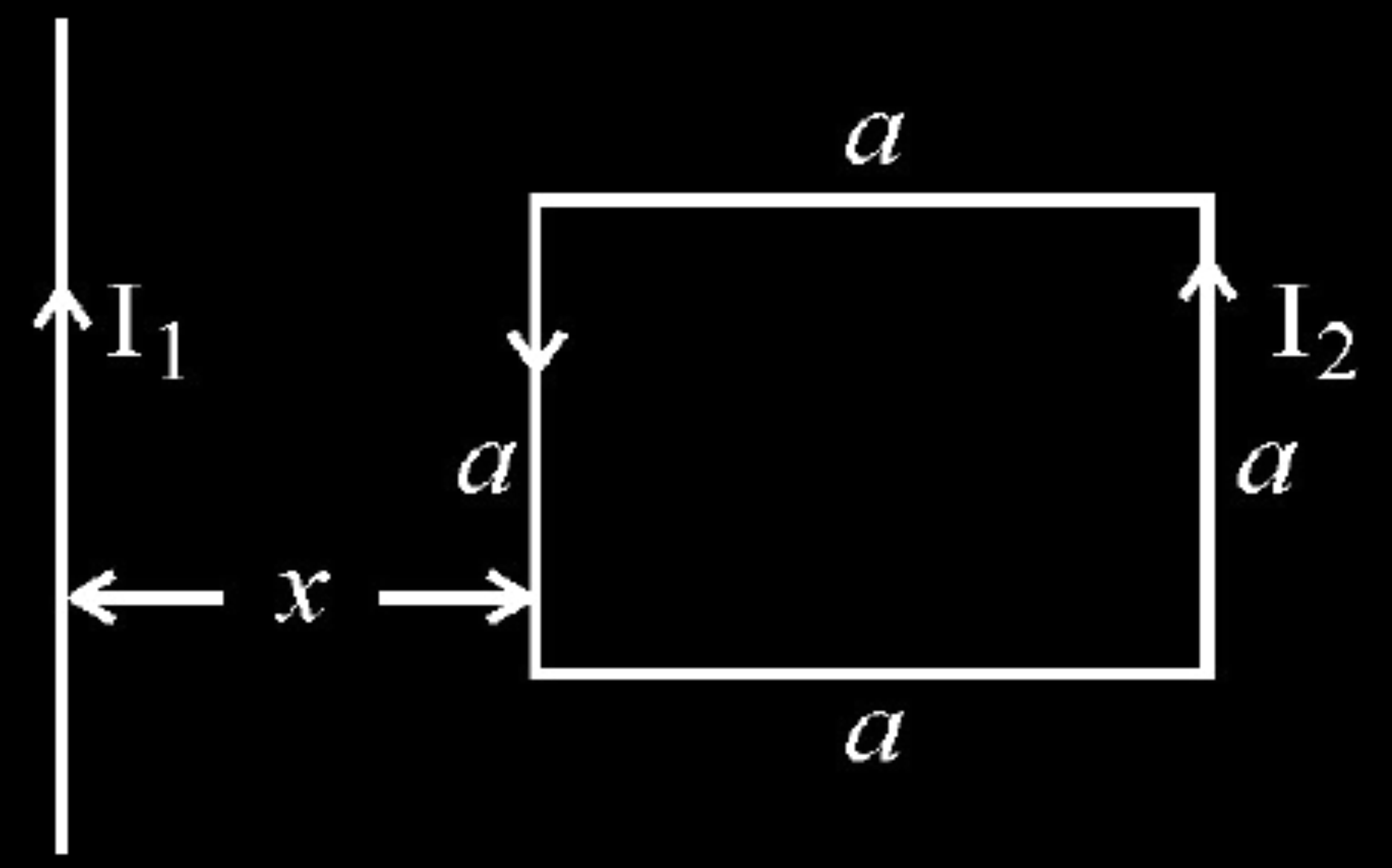
$$F = 7\times10^{-7}\,N$$

Net force is attractive in nature ½

As the lines of action of forces coincide torque is zero.



(b) A square loop of side 'a' carrying a current I_2 is kept at distance x from an infinitely long straight wire carrying a current I_1 as shown in the figure. Obtain the expression for the resultant force acting on the loop.



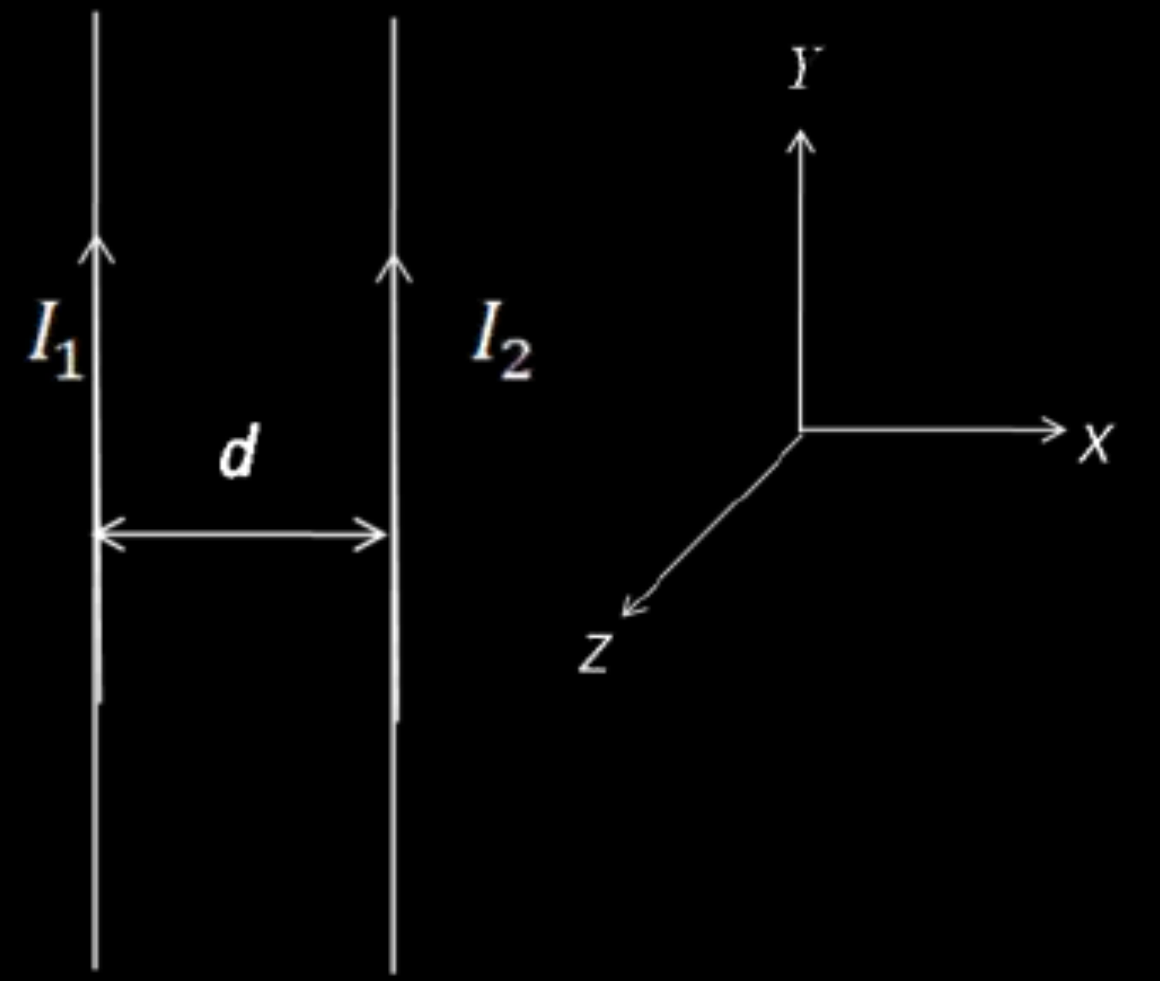
(b) Force per unit length between two parallel straight conductors $$F = \frac{\mu_0}{4\pi}\frac{2I_1I_2}{d}$$ Force on the part of the loop which is parallel to infinte straight wire and at a distance x from it. $$F_1 = \frac{\mu_0}{2\pi}\frac{I_1I_2\ a}{x}$$ (away from the infinte straight wire)	½
Force on the part of the loop which is at a distance (x + a) from it $$F_2 = \frac{\mu_0}{2\pi}\frac{I_1I_2\ a}{(x + a)}$$ (towards the infinte straight wire)	½
Net force $F = F_1 - F_2$ $$F = \frac{\mu_0}{2\pi}\frac{I_1I_2\ a}{}\left[\frac{1}{x} - \frac{1}{x+a}\right]$$ $$F = \frac{\mu_0}{2\pi}\frac{I_1I_2\ a^2}{x\ (x + a)}$$ (away from the infinte straight wire)	½

(a) Define SI unit of current in terms of the force between two parallel current carrying conductors.

(b) Two long straight parallel conductors carrying steady currents I_a and I_b along the same direction are separated by a distance d. How does one explain the force of attraction between them ? If a third conductor carrying a current I_c in the opposite direction is placed just in the middle of these conductors, find the resultant force acting on the third conductor.

Two long straight parallel conductors carry steady current I_1 and I_2 separated by a distance d. If the currents are flowing in the same direction, show how the magnetic field set up in one produces an attractive force on the other. Obtain the expression for this force. Hence define one ampere.

Diagram showing attractive force on other wire.	1
Obtaining an expression for force.	1
Definition of one ampere.	1

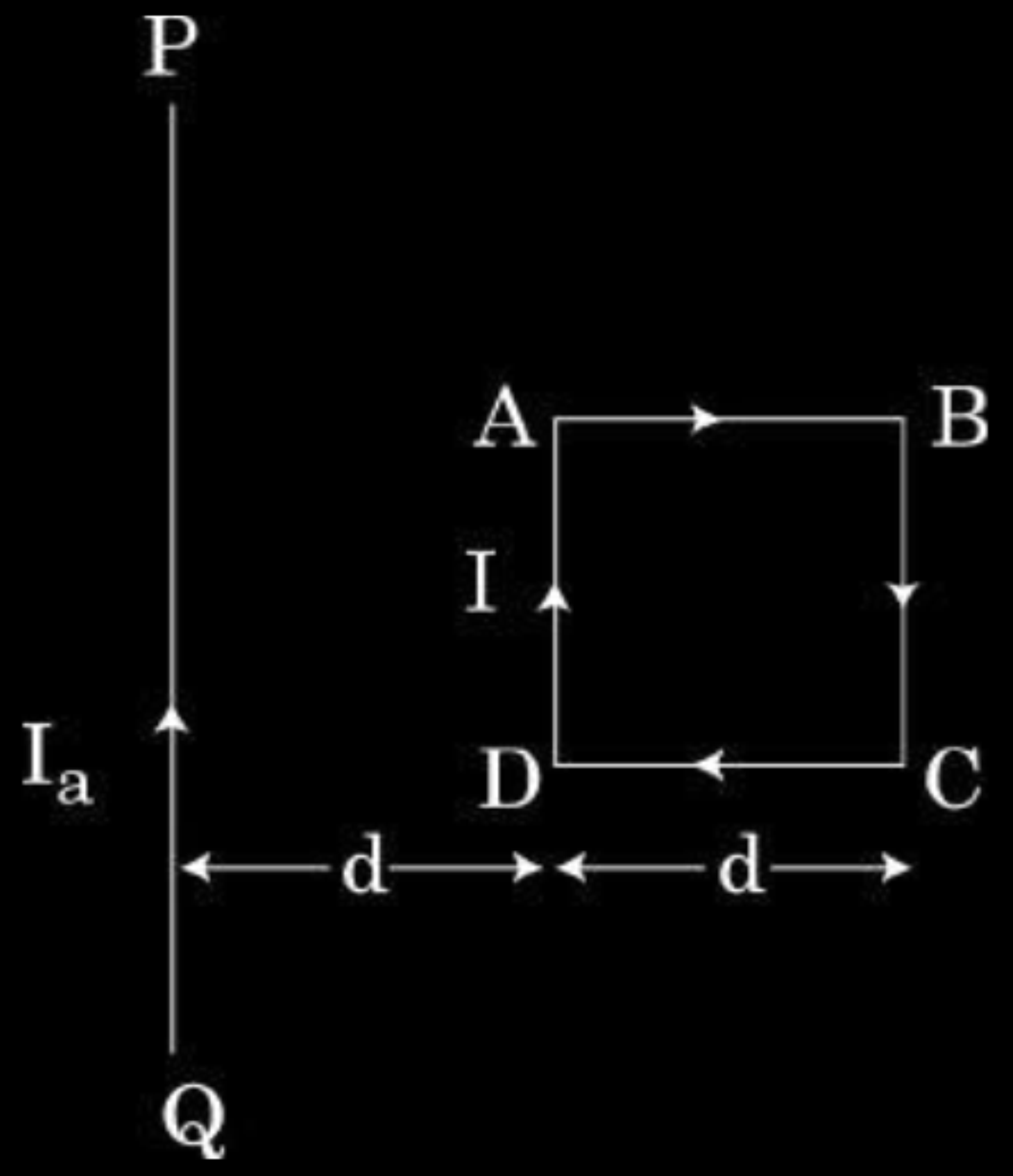
As shown in Figure, the direction of force on conductor b is attractive	½
[Alternatively: $\vec{B}$ at a point on wire 2, is along - $\widehat{k}$ $\therefore \vec{F}$, on wire 2, due to the $\vec{B}$, is along - $\hat{\imath}$, i.e. towards wire1. Hence the force is attractive. 	
Magnetic field, due to current in conductor a, $$B_1 = \frac{\mu_0 I_1}{2\pi d}$$	½
The magnitude of force on a length L of conductor b, $$F_2 = I_2 L B_1$$	½
$$F_2 = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$	
One ampere is that steady current which, when maintained in each of the two very long, straight, parallel conductors, placed one meter apart in vacuum, would produce on each of these conductors a force equal to 2×10^{-7} newton per meter of their length.	1

(i) Depict magnetic field lines due to two straight, long, parallel conductors carrying steady currents I_1 and I_2 in the same direction.

(ii) Write the expression for the magnetic field produced by one of the conductor over the other. Deduce an expression for the force per unit length.

(iii) Determine the direction of this force.

(iv) In figure given below, wire PQ is fixed while the square loop ABCD is free to move under the influence of currents flowing in them. State with reason, in which direction does the loop begin to move or rotate ?



(ii) $B_1 = \frac{\mu_0 I_1}{2\pi d}$ or $\frac{\mu_0 I_2}{2\pi d} = B_2$ ½

½

$$F = F_{12} = F_{21} = I_1 B_2 L = I_2 B_1 L$$

$$= \frac{\mu_0 I_1 I_2}{2\pi d}\Big) L$$

Force per unit length

$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$ ½

(iii) Attractive force ½

(iv) Loop ABCD will move towards wire PQ. ½

Current in wire PQ and Current in arm AD are in the same direction, so they attract each other. ½

Current in wire PQ and Current in arm BC are in opposite direction, so they repel each other. ½

Contribution due to current in AB and CD nullify each other.

Since arm AD is nearer than arm BC to arm PQ, so net force on the loop is attractive. Therefore, the loop will move towards the wire PQ. ½